



A Multi-Zone Unsteady Heat Transfer Model for an Open Volumetric Air Receiver: A Step Towards Scale-Up and Design Optimization

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ABSTRACT

A comprehensive, multi-zone, unsteady heat transfer model is developed for a straight absorber pore-based open volumetric air receiver. This includes heat exchange between the porous absorbers, absorbers and receiver casing, and absorbers and return air. The validation revealed its predictive capability within an uncertainty of $\pm 7\%$. The model is used for the scale-up of receiver design with several intermediate absorber layers. The major recommendations for scale-up are a) multiple absorber layers is beneficial for mitigating thermal stress, b) higher flux concentration is required for non-volumetric heating, to achieve the desired air temperature, compared to volumetric heating, c) air return ratio should be 0.6, d) absorber porosity should be 0.6 for volumetric heating and higher for non-volumetric heating, e) absorber gap to length ratio should be 0.15–0.25, f) the radiative heat loss is substantial for non-volumetric heating; therefore, the exposed surface area to ambient should be reduced, and g) absorber diameter to length ratio should be 1–2. The developed approach is generic and adaptable for the different open volumetric air receiver designs.

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1. Introduction

Solar thermal energy (STE) is one of the promising alternatives for net-zero carbon emissions. Solar tower systems are emerging as an efficient and viable option to harness the STE. Central receiver-based solar tower systems generate hot air at a temperature $> 700^\circ\text{C}$ using the open volumetric air receiver (OVAR) [1–3]. The OVAR comprises porous absorbers that are open to the atmosphere and are exposed to a Gaussian-type concentrated solar irradiance by a heliostat field. Air flowing through the OVAR is raised to a high temperature by the concentrated solar irradiance. The hot air can be used directly for an application or indirectly to heat a separate working fluid. The design of the porous absorbers affects the thermal efficiency of an OVAR and in turn the overall efficiency of the solar tower system [2]. Different absorber designs, like open-cell type, foam-type, and monolithic honeycombs have been in-

vestigated [4–8]. A recent literature review reported some of the critical challenges associated with the absorber design e.g. flow instability, thermal stress generation, and dust deposition [9,10]. A summary of various absorber designs is presented in Table 1. Experimental investigations [11–18] showed that the OVAR thermal efficiency depends on the design parameters like absorber geometry, absorber porosity, the mass flow rate of air, air return ratio (ARR), heat flux distribution, flux concentration, and the overall receiver design. However, the experimental investigations using large-scale solar towers are time and cost intensive, and are thus limited. Therefore, the mathematical or computational models, to improve and optimize the OVAR design, have been used extensively.

A large body of computational fluid dynamics (CFD) simulations can be found with varying degrees of complexity. 3D steady-state CFD investigations using FLUENT, including the Monte Carlo ray tracing (MCRT) method, are performed for volumetric and non-volumetric heating of OVAR [21]. This demonstrated a substantial impact of the heat flux distribution on the absorber temperature. Experiment and CFD revealed a substantial increase in the ARR for cavity, semi-cavity, and secondary-concentrator-type OVAR [22–25]. The OVAR with gradually varying porosity is studied us-

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Nomenclature

A	Area, m ²
K	Loss coefficient, Pa/m
C	central absorber
C_p	Specific heat, J/kg-K
d_{pr}	Diameter of pore, m
d_{ab}	Diameter of absorber, m
dh	Hydraulic diameter, m
D_R	Diameter of a ring, m
D_{cas}	Diameter of the casing, m
F	View factor
G	Gap, m
h	Convective heat transfer coefficient, W/m ² -K
I	Irradiance, W/m ²
k	Thermal conductivity, W/m-K
L	Length, m
M	intermediate absorber
$\dot{m}$	Mass flow rate, kg/s
np	Number of pores
N_{ab}	Total number of absorbers
N_R	Number of the absorber in ring
Nu	Nusselt number
P	peripheral absorber
Pr	Prandtl number
$\dot{q}$	Power, W
$\dot{q}_{cond}$	Rate of heat transfer due to conduction, W
$\dot{q}_{conv}$	Rate of heat transfer due to convection, W
$\dot{q}_{rad}$	Rate of heat transfer due to radiation, W
$\dot{q}_{gen}$	Volumetric heat generation, W/m ³
q''_{in}	Heat flux, W/m ²
Q	Volume flow rate, m ³ /s
r	Radius, m
$Re_{d_{pr}}$	Reynolds number based on pore diameter
Ra	Rayleigh number
T	Temperature, K
T_{out}	Average primary air outlet temperature, K
t	Time, s
V	Volume, m ³
z	Axial length, m

Subscript

a	Air
a_C	primary air in the central absorber
a_M	primary air in the intermediate absorber
a_P	primary air in the peripheral absorber
ab	Absorber
atm	Ambient
C	central absorber
cas	Casing
$cas-pu$	casing to peripheral absorber
dh	hydraulic diameter
$front$	Front
in	Inlet
M	intermediate absorber
$mu-cu$	intermediate absorber to central absorber
n	number of layers
nat	natural
out	Outlet
P	peripheral absorber
pr	Pore
$pu-mu$	peripheral to intermediate
R	Ring
r	Return

rad	radiation
rec	Receiver
rC	return air through the central region
rM	return air through the intermediate region
rP	return air through the peripheral region
st	Storage
th	Thermal

Greek

ε	Porosity
ϵ	Emissivity
β	extinction coefficient
σ	Variance
ρ	Density
μ	dynamic viscosity
θ	azimuthal angle
η	Efficiency

Abbreviation

ARR	Air Return Ratio
DNI	Direct Normal Irradiance
mfr	Mass Flow Rate
OVAR	Open Volumetric Air Receiver
POA	Power On Aperture

ing the 3D steady-state CFD simulations [26], and higher thermal efficiency is reported as compared to a constant porosity absorber. This study also reported the coefficients, for estimating the penetration depth, for the multi-layer absorbers. 3D CFD study for the performance estimation of OVAR and its comparison with experimental data for solar tower Jülich is reported in [27]. The study used appropriate boundary conditions to reduce the computational mesh size. For the steady-state heat transfer, the thermal efficiency of 70.9% at 650 °C and thermal efficiency of 75.4% at 450 °C is reported. A 3D CFD model including MCRT, for the thermal-hydraulic analysis of a porous absorber [28], suggested the use of high porosity. The CFD simulations are also performed for the quarter model of an OVAR with a glass cover to study the swirling flow through a varying porous structure [29]. The study reported thermal efficiency of 87% and recommended the use of a layered absorber design. Optical and thermal analysis of OVAR, by using 3D CFD with the radiation-based source term [30], found that the coated alumina honeycomb absorber reduces the reflection losses. This increases the outlet air temperature and thermal efficiency of an OVAR. Several studies have used the local thermal non-equilibrium (LTNE) model for investigating the heat-transfer processes inside a porous receiver. For example, the MCRT and LTNE -based 3D CFD model is used for the steady-state heat transfer analyses of OVAR in [31]. The work recommended an absorber with high porosity and small pore size for flux homogenization and high thermal efficiency. A 3D LTNE-based steady-state CFD heat transfer model, including heat-flux splitting, was validated with experiments [7,8,40]. The simulations suggested the use of peripheral heating to reciprocate the volumetric-type heating effect for the absorbers with high thermal conductivity. The 3D CFD simulations are valuable in terms of the insights they provide without requiring large-scale experimental setups. However, in the OVAR there is a wide separation of length scales, ~1 mm for the pore size to ~10 m for the receiver size, which leads to computationally challenging mesh size [27]. Thus, the computational requirement, for 3D simulations in OVAR, impedes its scale-up.

For reducing the computational cost, 2D CFD studies for OVAR, taking advantage of the problem symmetry and physics are also reported. A 2D steady-state CFD analysis for the maximum heat flux

Table 1

The porous absorber designs, material, and operating conditions for the OVARs.

Researcher	OVAR Design	Material	Outlet Temperature (°C)
Böhmer et al. [11]	Foil type	SiSiC	800
Hoffschmidt et al. [5]	Honeycomb (Hexagonal)	SiC	840
Sharma et al. [8]	Circular (straight pores)	Brass	350-360
Avila-Marin et al. [12]	Wire mesh (Gradual porosity)	Alloy	650
Pabst et al. [13]	Honeycomb (Hexagonal)	Metal	700-800
Avila-Marin et al. [14]	Graded mesh	Metal	>780
Du et al. [15,16]	Radially graded (Foam)	SiC	>660
Li et al. [19]	Honeycomb	Ceramic	550-575
Heisel et al. [20]	Various 3-D printed	SiC	860

of 2200 kW/m², performed with a SiC absorber having a porosity of 49.5% and wire-mesh alloy absorbers having a porosity of 97.4%, showed a deviation of about 10% with experiments [33]. The alloy-based absorber showed a greater thermal homogeneity in comparison to the SiC absorber, which is attributed to the higher porosity. A 2D, coupled, heat transfer model with MCRT is used to study the windowed volumetric receiver [34]. The LTNE effects are also considered in this work, and the thermal efficiency is investigated by varying cell size and porosity range.

The 1D simulations for OVARs are widely reported in the literature and are useful, as reliable results can be obtained with the least computational effort. Numerical investigation, including the 1D radiation transport equation (RTE), demonstrated the conditional airflow instability in OVAR [35] and concluded that the volumetric heating for an absorber may be achieved with an appropriate design. 1D analysis revealed the suppressed flow stability for the brass-based circular absorbers with straight pores [8,36,37]. The 1D analysis also revealed that the airflow resistance increases substantially with temperature and limits the concentrated solar irradiance onto porous absorbers [38]. Ray-tracing-based solar irradiance and radiation-based heat loss, in the 1D approach, are also developed for the thermal performance assessment of OVAR [39]. A direct positive correlation between foam-type absorber porosity and thermal efficiency of OVAR is developed using a 1D numerical model [40]. A coupled 1D heat transfer model was developed, including radiation exchange from the exposed absorber surface to the ambient, for the assessment of OVAR [41]. The study reported a high thermal efficiency for low inlet velocity and high thermal conductivity of the absorber material. A 1D numerical heat transfer model was presented for a solar receiver with a variable aperture mechanism in [42]. Various numerical models, developed at the German Aerospace Center (DLR), accounting for both the detailed and homogeneous representation of the absorber, are presented in [43]. This study reported a discrepancy between experimental and numerical results for temperature profiles in solid and fluid phases, as well as the efficiency. The literature survey revealed the predictive capability of the 1D models and underlines the need for a comprehensive, unsteady, heat transfer model benefiting from the insight and limitations of the unsteady state (1D) and steady-state (CFD) models, including all the modes of heat exchange a) between absorbers, b) between absorbers and return air, c) between absorber and receiver casing, and d) between absorbers and atmosphere. This is necessary for the scale-up, operation, and optimization of OVAR designs.

As a first step, a peripherally and centrally located absorbers-based two-zone, unsteady, heat transfer model was developed for an OVAR [44]. The novelty of the current work is the development of a multi-zone model for OVARs with several intermediate absorber layers, in addition to central and peripheral absorbers. To the best of the author's knowledge, such a detailed model is not yet reported for a realistic OVAR design. The framed objectives of the present work are as follows:

- Development of a multi-zone, unsteady, heat transfer model for OVAR including all the modes of heat exchange between absorbers, absorbers and return air, absorbers and receiver casing. The model also considers the radially-axially varying heat flux on the porous absorbers.
- Parametric investigation of the OVAR designs, with several absorber layers, to identify the important parameters for scale-up.

The paper is organized as follows: Section 2 presents the mathematical model, Section 3 presents results and discussions, and finally, the conclusions and recommendations are presented in Section 4.

2. Mathematical Modeling

In this section, a) the layout of scale-up OVAR designs with multiple intermediate absorber layers is presented, and b) the governing equations are deduced, for different receiver internals and airflow paths, along with the corresponding boundary conditions. The developed model will enable the scale-up of an OVAR design for a given operating condition.

2.1. Problem Description

A large field of heliostats is used to concentrate the direct solar radiation onto an OVAR, which comprises several absorber layers (see Fig. 1a). In this schematic, the absorbers are identified as Central (C), intermediate (M), and peripheral (P). Because of Gaussian-type flux distribution on the OVAR aperture, the central absorbers are exposed to the highest heat flux, intermediate absorbers are exposed to the intermediate heat flux, and the peripheral absorbers are exposed to the lowest heat flux. Therefore, the reported two-zone model [44] is extended, including the intermediate absorber layers, and is termed the multi-zone model. This model assumes a) incompressible airflow, b) absorbers of uniform length and porosity, and c) straight absorber pores. The straight absorber pore design was preferred for mitigating the serious problem of dust deposition in an absorber with complex porous structures [45,46]. The model requires, for instance, the absorber diameter (d_{ab}), the absorber length (L), the number of absorbers (N_R), the ring diameter (D_R), and the gap between absorbers (G) (see Fig. 1b). Here, the ring diameter for the first absorber layer is $D_{R1} = 2d_{ab} + 2G$ and for the n^{th} absorber layer is $D_{Rn} = D_{Rn-1} + (2d_{ab} + 2G)$ (see Fig. 1c). The number of absorbers for the n^{th} layer, $N_{Rn} = \pi D_{Rn} / (d_{ab} + G) - 1$, is used for the symmetry with an absorber azimuthal location $\theta_R = 360^\circ / N_{Rn}$ (see Fig. 1d and 1e). The inner diameter of the receiver casing is given by $D_{cas} = D_{Rn-1} + G + d_{ab}$. All the absorbers are assumed to have the same number of pores (np) with diameter d_{pr} . The developed model offers flexibility to these parameters, and thus adaptable for the different OVAR designs. The z-direction of airflow, inside and outside of the absorbers, is depicted using solid and dotted arrows

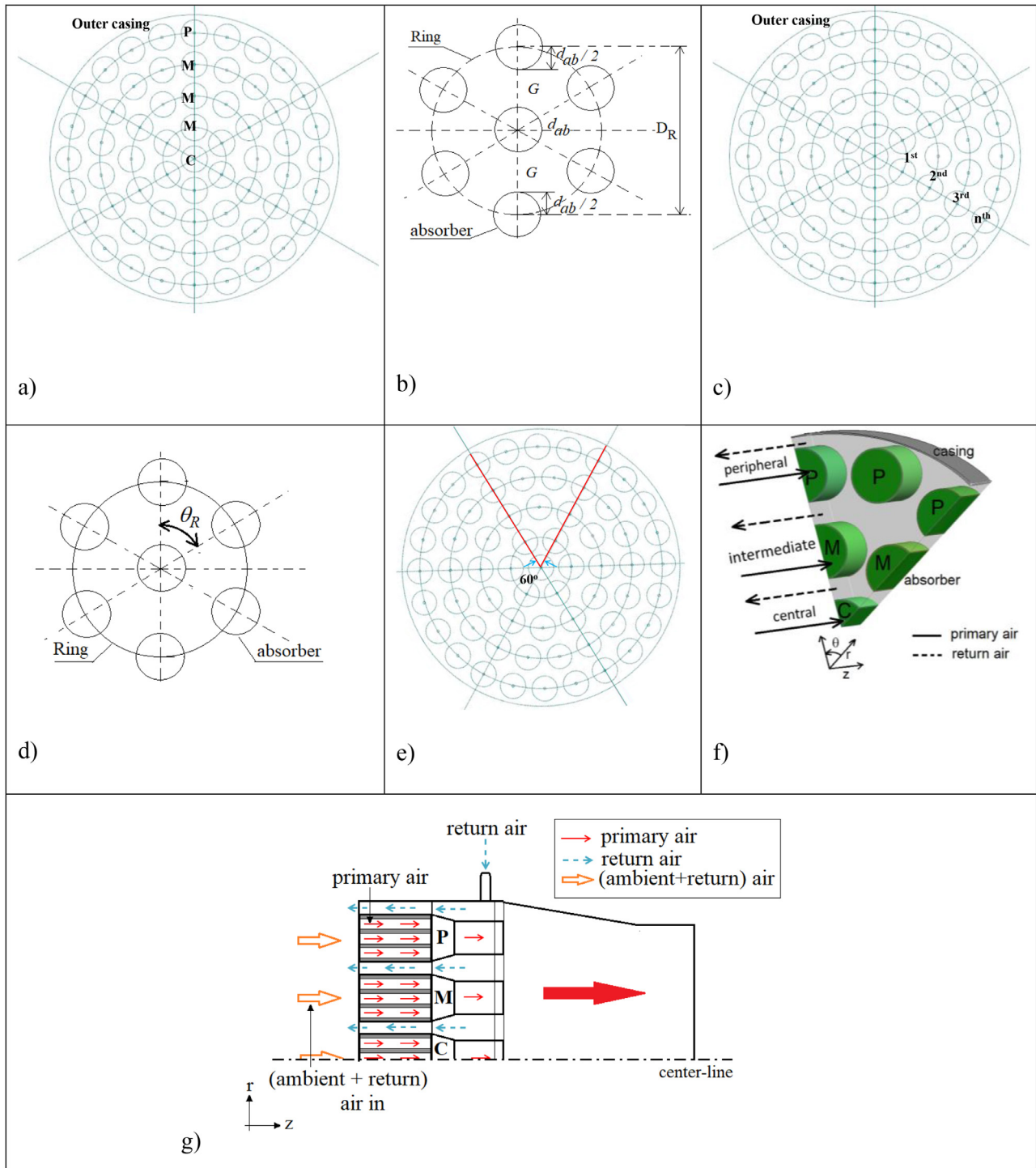


Fig. 1. The schematic showing the a) circular arrangement, b) absorbers in the first ring, c) location of the rings, d) the angle between two absorbers, e) symmetry, f and g) the path and direction for primary airflow inside an absorber and the return airflow around the absorber.

(see Fig. 1f and g). These figures depict the flow path of primary air, which is a mixture of return air and ambient air, and return air around the absorbers. It may be noted that the direction of primary air and return air flow are opposite to each other. The hot primary air at the OVAR outlet flows through thermal energy storage or an application. There, the heat from primary air is recovered and the relatively cooled, return air, enters the OVAR for external cooling and temperature redistribution of absorbers. The hot air from the absorber outlet flows through the nozzle-shaped mixture

assembly to provide a uniform, high-temperature, hot air at the OVAR outlet [7,8,32]. The entire OVAR assembly is well-insulated, externally, for mitigating heat loss to the ambient.

The literature review reveals that the porous absorbers are heated volumetrically or non-volumetrically. In-house experiments and CFD simulations revealed that the volumetric (uniform) absorption of heat in the absorbers promotes the desired volumetric heating for an OVAR [7,8,32]. Hence, internal heating of absorbers is applied for simulating volumetric heating (see Fig. 2a). This led

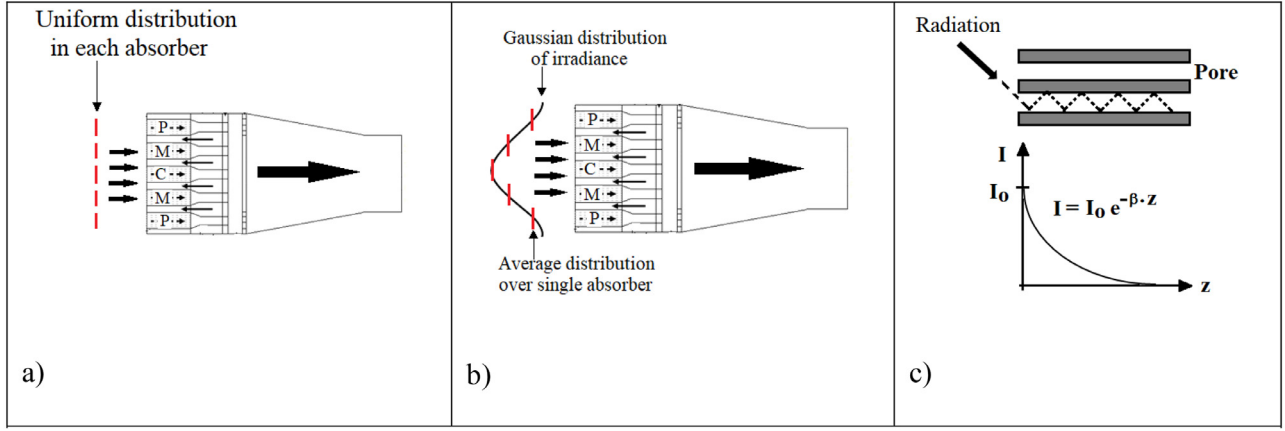


Fig. 2. Schematic showing (a) volumetric heating of an absorber, (b, c) Gaussian-type irradiance onto an absorber, and heat flux variation inside of an absorber for non-volumetric heating.

to the lowest absorber surface temperature at the inlet, $z/L = 0$ and the highest absorber surface temperature at the outlet, $z/L = 1$ [62]. The heliostat field concentrates the beam solar radiation onto the OVAR aperture. A part of this concentrated irradiance, falling onto the front surface of an absorber, follows a non-homogeneous, radial, Gaussian-type distribution [47,48]. The remaining part enters through the absorber pore and is internally absorbed. Therefore, the non-volumetric heating is simulated by (a) Gaussian-type irradiance onto the front surface of an absorber is shown in Fig. 2b, and (b) the decreasing flux inside of an absorber pore, along the z -direction, is shown in Fig. 2c. This leads to the highest absorber temperature at the inlet, $z/L = 0$ and the lowest absorber surface temperature at the outlet, $z/L = 1$. Therefore, non-volumetric heating leads to a higher heat loss from the absorber surface compared to volumetric heating. The radial and axial variations of irradiance are calculated as follows [6]:

$$I(r_{rec}) = I_0 \cdot \exp\left(\frac{-1}{2} \cdot \left(\frac{r_{rec}^2}{\sigma^2}\right)\right) \quad (1)$$

$$I(z) = I_0 e^{-\beta \cdot z} \quad (2)$$

Where I_0 is the maximum heat flux at $r_{rec} = 0$, $\beta = 3(1-\varepsilon)/d_{pr}$ [49] is the estimated extinction coefficient, ε is the absorber porosity, and d_{pr} is the pore diameter for an absorber. The multi-zone model includes several intermediate absorber layers between central and peripheral absorbers. The absorber layouts for one, two, and three intermediate absorber layers are illustrated in Fig. 3. The total number of absorbers N_{ab} in an OVAR is 19 for the one intermediate absorber layer and 61 for the three intermediate absorber layers. The actual number of intermediate absorber layers will depend upon the capacity of an OVAR design.

To scale up an OVAR design, with multiple intermediate absorber layers and for a given air mass flow rate ($\dot{m}_a$), the absorber diameter, the absorber porosity, and the Reynolds number based on pore diameter are maintained. This allows a comparative assessment between the different OVAR designs for optimization. The air mass flow rate through one absorber will be $\dot{m}_{ab} = \dot{m}_a/N_{ab}$, the air mass flow rate through a single pore is $\dot{m}_{pr} = \dot{m}_{ab}/np$, where np is the total number of pores in an absorber. Thus, the Reynolds number based on pore diameter $Re_{d_{pr}}$ is given by,

$$Re_{d_{pr}} = \frac{4 \dot{m}_a}{\pi d_{pr} \mu N_{ab} np} \quad (3)$$

Here, the pore diameter (d_{pr}) is to be selected to obtain the desired $Re_{d_{pr}}$ and is given by,

$$d_{pr} = \frac{4}{\pi \mu Re_{d_{pr}}} \frac{\dot{m}_a}{N_{ab} np} \quad (4)$$

Now, for a given absorber porosity (ε) the absorber diameter is given by,

$$d_{ab} = \left[\frac{1}{\varepsilon np} \left(\frac{4}{\pi \mu Re_{d_{pr}}} \frac{\dot{m}_a}{N_{ab}} \right)^2 \right]^{1/2} \quad (5)$$

Eq. (5) provides the required absorber diameter for different intermediate absorber layers.

2.2. Governing equations

The differential form of the governing equations is derived for the receiver casing, the central absorbers, the intermediate absorbers, the peripheral absorbers, and the return air side, considering the 60° symmetry of an OVAR design (see Fig. 4a). A differential element (dz) along the main flow (z) direction is shown in Fig. 4b. Here, the absorbers are highlighted as C, M, and P depending on their relative positions in an OVAR. All the modes of heat transfer, including the interaction between different receiver internals, are depicted using Fig. 4c. This reveals the involved complex heat transfer processes, which are mathematically modeled, and the same is presented subsequently. In this paper, one intermediate absorber layer is used to illustrate the model development. For several intermediate absorber layers, the corresponding equation is to be solved repeatedly with appropriate view factors.

2.2.1. Receiver casing

The receiver casing encloses all the porous absorbers and is externally insulated. This arrangement mitigates heat loss from the circumference of the absorber to the ambient. The heat transfer modeling of the receiver casing excludes a) the direct solar irradiance and b) the direct radiation exchanges with the central (C) and intermediate (M) absorbers on account of the small gap between absorbers. The model includes the radiative and convective heat exchanges with the peripheral absorbers (P) and the return air (see Fig. 5). The return air flows through the space between the receiver casing and P. In this figure, $\dot{q}_{conv_rp}$ denotes the rate of convective heat transfer from casing to return air, $\dot{q}_{cond_z}$ denotes the rate of heat conduction in the receiver casing at z , and $\dot{q}_{rad\ (cas \leftrightarrow P)}$ denotes the net radiative heat exchange between the receiver casing and P.

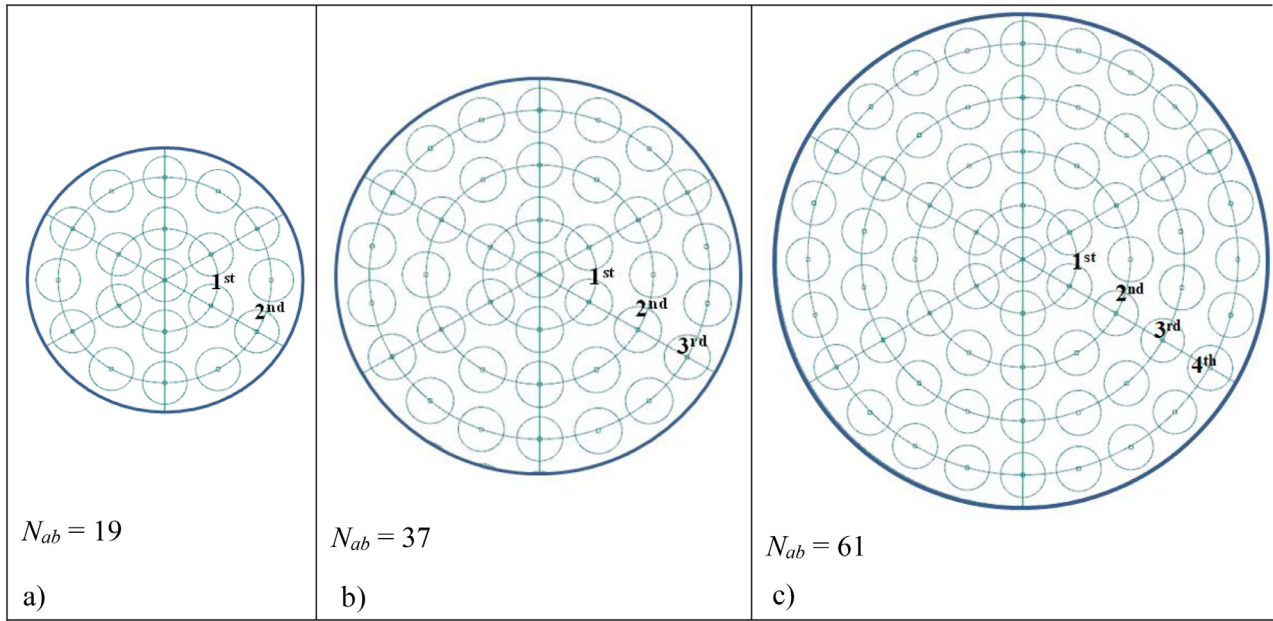


Fig. 3. Front view of an open volumetric air receiver with a) one, b) two, and c) three intermediate absorber layers.

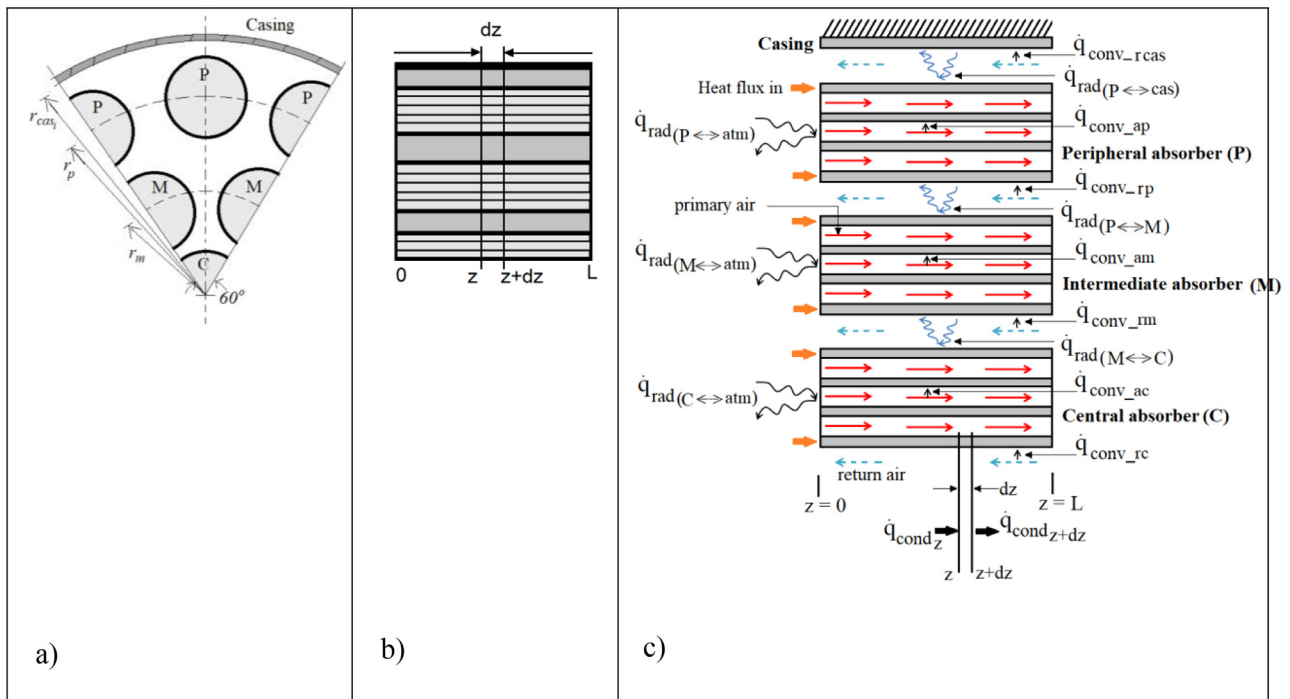


Fig. 4. A schematic showing, a) 60° symmetry of an OVAR design and b) the differential element along the z-direction and c) all the modes of heat transfer in absorber.

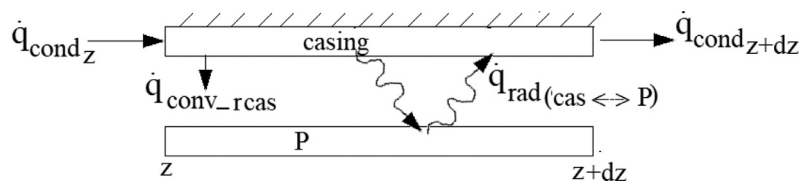


Fig. 5. A schematic showing the heat exchanges between the receiver casing, and the peripheral absorbers and the return air.

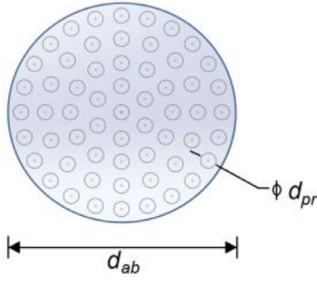


Fig. 6. A schematic showing the cross-section of a porous absorber and the important geometrical parameters.

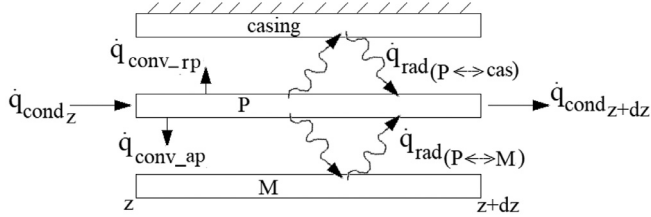


Fig. 7. A schematic showing the different modes of heat transfer to and from the peripheral absorbers (P).

The instantaneous energy balance, $\dot{q}_{in} - \dot{q}_{out} = \dot{q}_{st}$, results in the following expression:

$$\underbrace{\frac{k_{cas}}{\rho_{cas} c p_{cas}} \frac{\partial^2 T_{cas}}{\partial z^2}}_{\text{conduction term}} - \underbrace{\frac{2h_{rp} r_{casin}}{\rho_{cas} c p_{cas} (r_{casout}^2 - r_{casin}^2)} (T_{cas} - T_{rp})}_{\text{convection term}} + \underbrace{\frac{\sigma (T_p^4 - T_{cas}^4)}{\rho_{cas} c p_{cas} \left[\left(\frac{1-\epsilon_{cas}}{\epsilon_{cas}} \right) \frac{3(r_{casout}^2 - r_{casin}^2)}{r_{casin}} + \frac{3(r_{casout}^2 - r_{casin}^2)}{r_{casin} F_{cas-pu}} + \left(\frac{1-\epsilon_p}{\epsilon_p} \right) \frac{(r_{casout}^2 - r_{casin}^2)}{2r_{ab}} \right]}}_{\text{radiation exchange term}} = \underbrace{\frac{\partial T_{cas}}{\partial t}}_{\text{storage term}} \quad (6)$$

In Eq. (6) the conduction, convection, and radiation exchange related terms are identified for the receiver casing. The prescribed initial condition (at $t = 0$) is $T_{cas} = T_{atm}$. The prescribed boundary conditions at the inlet and outlet are as follows:

$$\text{Inlet (at } z = 0): k_{cas} \frac{\partial T_{cas}}{\partial z} = \sigma \epsilon_{cas} (T_{cas}^4 - T_{atm}^4)$$

$$\text{Outlet (at } z = L): \frac{\partial T_{cas}}{\partial z} = 0.$$

2.2.2. Absorber

A porous absorber is depicted using the schematic in Fig. 6. The important geometrical parameters like pore diameter, and absorber diameter, are depicted in this figure. Depending on the locations, the absorbers in an OVAR are identified as peripheral (P), intermediate (M), and central (C) in Fig. 4. The unsteady heat transfer modeling for these absorbers is described subsequently. The primary air flows inside the pores of these absorbers. The return air flows around the absorber in between the narrow gaps.

2.2.2.1. Peripheral absorber. The modes of heat transfer to and from P are depicted in Fig. 7. These absorbers interact with the receiver casing, M, and the return air around these absorbers. The rate of heat transfer, via forced convection, from P to primary air is given by $\dot{q}_{conv-ap}$, the rate of heat transfer, via forced convection, from P to return air is given by $\dot{q}_{conv-rp}$, the rate of heat transfer due to conduction at z is given by $\dot{q}_{cond_z}$, the net radiation-based heat

exchange between P and the receiver casing is given by $\dot{q}_{rad(cas \leftrightarrow P)}$, and the net heat radiation-based heat exchange between P and M is given by $\dot{q}_{rad(M \leftrightarrow P)}$.

The instantaneous energy balance gives,

$$\underbrace{\frac{k_p}{\rho_p c p_p} \frac{\partial^2 T_p}{\partial z^2}}_{\text{conduction term}} - \underbrace{\frac{2h_{ap} n p r_{pr}}{\rho_p c p_p (1-\epsilon) r_{ab}^2} (T_p - T_{a,p})}_{\text{convection term from P to primary air}} - \underbrace{\frac{2h_{rp}}{\rho_p c p_p (1-\epsilon) r_{ab}} (T_p - T_{rp})}_{\text{convection term from P to return-air}} + \underbrace{\frac{\sigma (T_{cas}^4 - T_p^4)}{\rho_p c p_p \left[\left(\frac{1-\epsilon_p}{\epsilon_p} \right) (1-\epsilon) r_{ab} + \frac{(1-\epsilon) r_{ab}}{F_{pu-cas}} + \left(\frac{1-\epsilon_{cas}}{\epsilon_{cas}} \right) \frac{6(1-\epsilon) r_{ab}^2}{r_{cas,i}} \right]}}_{\text{radiation exchange term between P and casing}} + \underbrace{\frac{\sigma (T_M^4 - T_p^4)}{\rho_p c p_p \left[\left(\frac{1-\epsilon_p}{\epsilon_p} \right) (1-\epsilon) r_{ab} + \frac{(1-\epsilon) r_{ab}}{F_{pu-mu}} + \left(\frac{1-\epsilon_M}{\epsilon_M} \right) (1-\epsilon) 2r_{ab} \right]}}_{\text{radiation exchange term between P and M}} + \underbrace{\frac{\dot{q}_p}{\rho_p c p_p dV_p}}_{\text{heat input term}} = \underbrace{\frac{\partial T_p}{\partial t}}_{\text{unsteady term}} \quad (7)$$

Here, the heat source to P for volumetric and non-volumetric heating is given as follows:

$$\dot{q}_p = \begin{cases} \dot{q}_{gen} dV_p & \text{for volumetric heating} \\ q''_{in-p} dA_{p_s} & \text{for non-volumetric heating} \end{cases}$$

For simulating the volumetric heating, for instance using the peripheral heating in the laboratory [7,8], the uniform internal heat generation ($\dot{q}_{gen}$) is applied. For the non-volumetric heating, with concentrated solar radiation on the field, the heat flux q''_{in-p} (e.g. $I(r)$) is applied on the front surface and the distribution $I(z)$ is applied along the axial direction. The prescribed initial and boundary conditions are as follows:

Initial condition: The temperature of the peripheral absorber is equal to the ambient, i.e., at $t = 0$ $T_p = T_{atm}$.

Boundary conditions: The boundary conditions, for volumetric and non-volumetric heating, are applied at the inlet ($z = 0$) and the outlet ($z = L$) to the absorber. These are as follows: a) at $z=0$:

$$\underbrace{k_p \frac{\partial T_p}{\partial z}}_{\text{conduction term}} = \underbrace{\sigma \epsilon_p (T_p^4 - T_{atm}^4)}_{\text{radiation exchange between P and atmosphere}} + \underbrace{h(T_p - T_{atm})}_{\text{convection term}}$$

for the volumetric heating, and

$$\underbrace{-k \frac{\partial T_p}{\partial z}}_{\text{conduction term}} = \underbrace{q''_{in-p}}_{\text{input heat flux}} - \underbrace{\sigma \epsilon_p (T_p^4 - T_{atm}^4)}_{\text{radiation exchange between P and atmosphere}} - \underbrace{h(T_p - T_{atm})}_{\text{convection term}}$$

for the non-volumetric heating, and b) the outlet is treated as adiabatic, considering the externally insulated absorbers, i.e. at $z = L$: $\frac{\partial T_p}{\partial z} = 0$. The absorbers are internally (volumetrically) heated for simulating the volumetric heating effect. Therefore, the heat flux term is not included as the inlet boundary condition for volumetric heating.

2.2.2.2. Intermediate absorber. The multi-zone model includes the heat transfer in M, which is located in between P and C (Fig. 4). Thus, M interacts with P, C, and the return air. The direct radiation

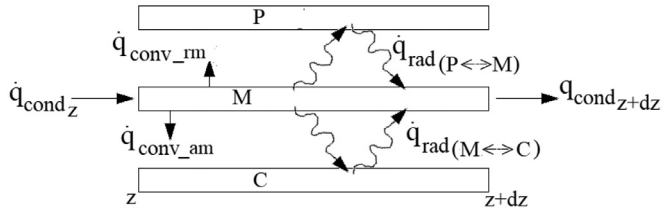


Fig. 8. A schematic showing the modes of heat transfer to and from an intermediate absorber (M).

exchange between M and receiver casing is neglected, assuming the small gaps between the absorbers. The different modes of heat exchange to and from M are depicted in Fig. 8. In this figure, the rate of heat transfer, via forced convection, from M to primary air is $\dot{q}_{conv_am}$, the rate of heat transfer, via forced convection, from M to the return air is $\dot{q}_{conv_rm}$, the rate of heat conduction in M at an axial position z is $\dot{q}_{cond_z}$, the net radiation-based heat exchange between M and P is $\dot{q}_{rad}(P \leftrightarrow M)$ and between M and C is $\dot{q}_{rad}(M \leftrightarrow C)$.

The instantaneous energy balance for M is given by,

$$\underbrace{\frac{k_M}{\rho_M c p_M} \frac{\partial^2 T_M}{\partial z^2}}_{\text{conduction term}} - \underbrace{\frac{2h_{ac} n_M r_{pr}}{\rho_M c p_M (1-\epsilon) r_{ab}^2} (T_M - T_{a_M})}_{\text{convection term from M to primary air}} - \underbrace{\frac{2h_{rc}}{\rho_M c p_M (1-\epsilon) r_{ab}} (T_M - T_{r_M})}_{\text{convection term from M to return-air}} + \underbrace{\frac{\sigma (T_P^4 - T_M^4)}{\rho_M c p_M \left[\left(\frac{1-\epsilon_M}{\epsilon_M} \right) (1-\epsilon) r_{ab} + \frac{(1-\epsilon) r_{abs}}{F_{mu-pu}} + \left(\frac{1-\epsilon_M}{\epsilon_M} \right) \frac{(1-\epsilon) r_{ab}}{2} \right]}}_{\text{radiation exchange term between P and M}} + \underbrace{\frac{\sigma (T_C^4 - T_M^4)}{\rho_M c p_M \left[\left(\frac{1-\epsilon_M}{\epsilon_M} \right) (1-\epsilon) r_{ab} + \frac{(1-\epsilon) r_{ab}}{F_{mu-cu}} + \left(\frac{1-\epsilon_M}{\epsilon_M} \right) 3(1-\epsilon) r_{ab} \right]}}_{\text{radiation exchange term between C and M}} + \underbrace{\frac{\dot{q}_M}{\rho_M c p_M dV_M}}_{\text{heat input term}} = \underbrace{\frac{\partial T_M}{\partial t}}_{\text{unsteady term}} \quad (8)$$

The heat source to M, for volumetric and non-volumetric heating, is as follows:

$$\dot{q}_M = \begin{cases} \dot{q}_{gen} dV_M & \text{for volumetric heating} \\ q''_{in-M} dA_{M_s} & \text{for non-volumetric heating} \end{cases}$$

Here, the volumetric heating is simulated using the uniform, internal, heat generation and the non-volumetric heating is simulated by using the radial and axial variation of heat-flux on the absorber. The prescribed initial and boundary conditions are as follows:

Initial condition: The temperature of the intermediate absorber is equal to ambient i.e. at $t = 0$, $T_M = T_{atm}$.

Boundary conditions: The prescribed boundary conditions are a) at the inlet ($z = 0$):

$$\underbrace{k_M \frac{\partial T_M}{\partial z}}_{\text{conduction term}} = \underbrace{\sigma \epsilon_M (T_M^4 - T_{atm}^4)}_{\text{radiation exchange between M and atmosphere}} + \underbrace{h(T_M - T_{atm})}_{\text{convection term}}$$

for volumetric heating and

$$\underbrace{-k_M \frac{\partial T_M}{\partial z}}_{\text{conduction term}} = \underbrace{q''_{in-M}}_{\text{input heat flux}} - \underbrace{\sigma \epsilon_M (T_M^4 - T_{atm}^4)}_{\text{radiation exchange between M and atmosphere}} - \underbrace{h(T_M - T_{atm})}_{\text{convective term}}$$

for non-volumetric heating, and b) at the outlet, considering the external insulation, the adiabatic boundary condition is applied i.e. at $z = L$, $\frac{\partial T_M}{\partial z} = 0$.

2.2.2.3. Central absorber. The absorber located at the center is termed the central absorber (C). This absorber exchanges heat with M, the primary air, and the return air. Fig. 9 shows the different modes of heat exchange for C. The rate of heat transfer, via forced convection, from C to primary air is given by $\dot{q}_{conv_ac}$, the rate of heat transfer, via forced convection, from C to return air is given by $\dot{q}_{conv_rc}$, the net radiation-based heat exchange between C and M is given by $\dot{q}_{rad}(C \leftrightarrow M)$. Here, the direct interaction between C and P is ignored considering the presence of M and small gaps between absorbers.

The instantaneous energy balance gives,

$$\underbrace{\frac{k_C}{\rho_C c p_C} \frac{\partial^2 T_{Cabs}}{\partial z^2}}_{\text{conduction term}} - \underbrace{\frac{2h_{ac} n_C r_{pr}}{\rho_C c p_C (1-\epsilon) r_{ab}^2} (T_C - T_{a_C})}_{\text{convection term from C to primary air}} - \underbrace{\frac{2h_{rc}}{\rho_C c p_C (1-\epsilon) r_{ab}} (T_C - T_{r_C})}_{\text{convection term from C to return-air}} + \underbrace{\frac{\sigma (T_C^4 - T_M^4)}{\rho_C c p_C \left[\left(\frac{1-\epsilon_C}{\epsilon_C} \right) \frac{(1-\epsilon) r_{ab}}{2} + \frac{(1-\epsilon) r_{ab}}{2F_{cu-mu}} + \left(\frac{1-\epsilon_C}{\epsilon_C} \right) \frac{(1-\epsilon) r_{ab}}{6} \right]}}_{\text{radiation exchange term between C and M}} + \underbrace{\frac{\dot{q}_C}{\rho_C c p_C dV_C}}_{\text{heat input term}} = \underbrace{\frac{\partial T_C}{\partial t}}_{\text{unsteady term}} \quad (9)$$

As in P and M, the heat input, for volumetric and non-volumetric heating, is as follows:

$$\text{Here, } \dot{q}_C = \begin{cases} \dot{q}_{gen} dV_C & \text{for volumetric heating} \\ q''_{in-C} dA_{C_s} & \text{for non-volumetric heating} \end{cases}$$

The prescribed initial and boundary conditions for solving this partial differential Eq. (9) are as follows:

Initial condition: The temperature of the central absorber is equal to ambient i.e. at $t = 0$, $T_C = T_{atm}$.

Boundary conditions: The boundary conditions are prescribed at the inlet and the outlet. These are as follows: a) at the inlet $z = 0$,

$$\underbrace{k_C \frac{\partial T_C}{\partial z}}_{\text{conduction term}} = \underbrace{\sigma \epsilon_C (T_C^4 - T_{atm}^4)}_{\text{radiation exchange between C and atmosphere}} + \underbrace{h(T_C - T_{atm})}_{\text{convection term}}$$

for volumetric heating and

$$\underbrace{-k_C \frac{\partial T_C}{\partial z}}_{\text{conduction term}} = \underbrace{q'_{in-C}}_{\text{input heat flux}} - \underbrace{\sigma \epsilon_C (T_C^4 - T_{atm}^4)}_{\text{radiation exchange between C and atmosphere}} - \underbrace{h(T_C - T_{atm})}_{\text{convection term}}$$

for non-volumetric heating, and b) at the outlet, considering external insulation, the adiabatic condition is applied, i.e. at $z = L$, $\frac{\partial T_C}{\partial z} = 0$.

2.2.3. Primary air

The primary air, a mixture of ambient and return air, flows inside the pores of an absorber. This leads to heat removal from the absorbers or heating of the primary air via forced convection. The heat transfer to primary air is identical for all the absorbers viz. P, M, and C. Therefore, one of the absorbers, namely P, is selected to

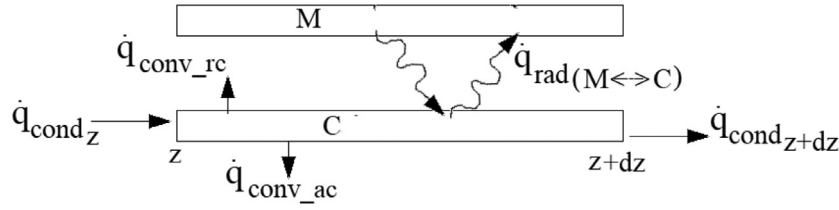


Fig. 9. A schematic showing the different modes of heat transfer to and from the central absorber (C).

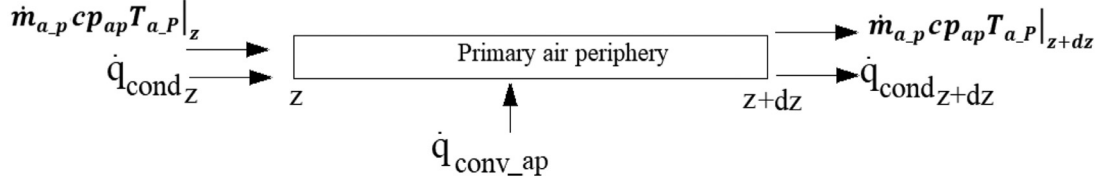


Fig. 10. Schematic showing heat transfer to and from primary air at peripheral absorber (P).

deduce the heat transfer model for primary air (see Fig. 10). The rate of heat transfer due to the advection of primary air at \$z\$ is given by \$\dot{m}_{a,p} c p_{ap} T_{a,p}|_z\$, the rate of heat transfer due to conduction at \$z\$ is given by \$\dot{q}_{\text{cond}_z}\$ and at \$z+dz\$ is \$\dot{q}_{\text{cond}_{z+dz}}\$, and the rate of heat transfer, via forced convection, from an absorber to primary air is given by \$\dot{q}_{\text{conv_ap}}\$.

The instantaneous energy balance, for primary airflow inside P, gives,

$$\underbrace{\frac{k_{ap}}{\rho_{ap} c p_{ap}} \frac{\partial^2 T_{a,p}}{\partial z^2}}_{\text{conduction term}} - \underbrace{\frac{\dot{m}_{a,p}}{2n\pi r_{pr}^2 \rho_{ap}} \frac{\partial T_{a,p}}{\partial z}}_{\text{advection term in primary air}} + \underbrace{\frac{2h_{a,p}}{\rho_{ap} c p_{ap} r_{pr}} (T_p - T_{a,p})}_{\text{convection term from P to primary air}} = \underbrace{\frac{\partial T_{a,p}}{\partial t}}_{\text{unsteady term}} \quad (10)$$

The identical equations can be deduced for primary airflow inside the pores of intermediate and central absorbers. Thus, the same is not repeated without the loss of generality. The prescribed initial and boundary condition for solving Eq. (10) are as follows:

Initial condition: The temperature of primary is equal to the ambient i.e. at \$t = 0\$; \$T_{a,p} = T_{atm}\$.

Boundary conditions: These are to be specified at the inlet and outlet to an absorber. During the operation of an OVAR the ambient air mixed with the return air at the inlet. Therefore, the conservation of energy provides at \$z = 0\$, \$T_{a,p} = ARR \times T_{rp}|_{z=0} + (1 - ARR)T_{atm}\$. At the outlet, considering external insulation, the adiabatic condition is applied, i.e. at \$z = L\$, \$\frac{\partial T_{a,p}}{\partial z} = 0\$. Generally, the experiments were used to estimate the ARR for an OVAR design see e.g. [5,22,50–52]. These investigations reveal that ARR depends on several parameters such as absorber and receiver design, air mass flow rate and temperature, and ambient conditions.

For identical designs of P, M and C, the heat transfer coefficient for primary airflow are estimated using the Reynolds number based on pore diameter (\$Re_{d_{pr}}\$) [53]:

$$Nu_a = 4.36 + \frac{0.086(Re_{d_{pr}} Pr_{pr}^{\frac{1}{3}})}{1 + 0.1Pr_{pr}^{\frac{1}{3}}} \quad (11)$$

For non-identical absorbers, the above expression can be easily adapted. Thus, the developed approach is generic, and the model is adaptable for the different OVAR designs.

2.2.4. Return air

The hot primary air is allowed to flow through thermal energy storage (TES) or an application, like solar convective furnace. Here

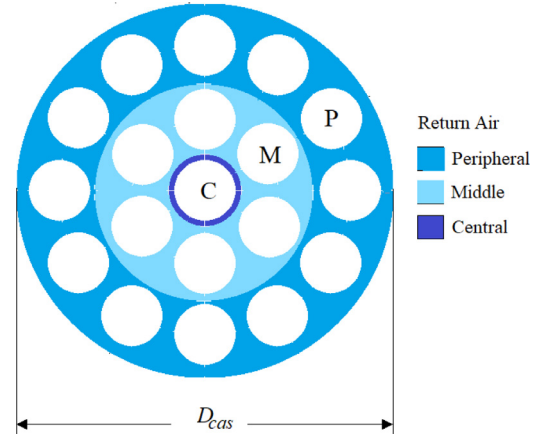


Fig. 11. Schematic showing the return airflow, through the different regions, for one intermediate absorber layer.

the energy from primary air is recovered, and the resulting, relatively cooled, return air is introduced around the C, M, and P through the return air flow chamber. The return air is used for the cooling and redistribution of heat among the absorbers for thermal uniformity. The direction of return airflow is opposite to that of the primary airflow (see Fig. 1f and g). For heat transfer modeling with return air, the identified central region, intermediate region and peripheral region, consistent with the absorbers, are depicted in Fig. 11. The return air flowing through the peripheral region interacts with the casing and P. The return air flowing through the intermediate region interacts with M. The return air flowing through the central region interacts with C. Thus, the identified mutually exclusive regions, using the smallest gap between absorbers as in [44], reduce the model complexity to some extent. The inlet return air temperature at \$z/L=1\$ is maintained at 300 K for all the simulations, except for the parametric investigation to analyze its effect on the performance of an OVAR. At the absorber inlet, return air coming out of the OVAR is mixed with the ambient air as shown in Fig. 1g. This serves as the primary air (P) for the OVAR, which flows through the pores of an absorber. A 3D CFD analysis for return airflow revealed the presence of cross-flow and vortex around the absorbers, which promotes the mixing of return air with ambient air, and the same was inferred from the experimental data [7,8]. The mixing of ambient and return air is modelled with the inlet boundary condition for the primary air and is described in the Section 2.2.3. For an incompressible flow of return air in these

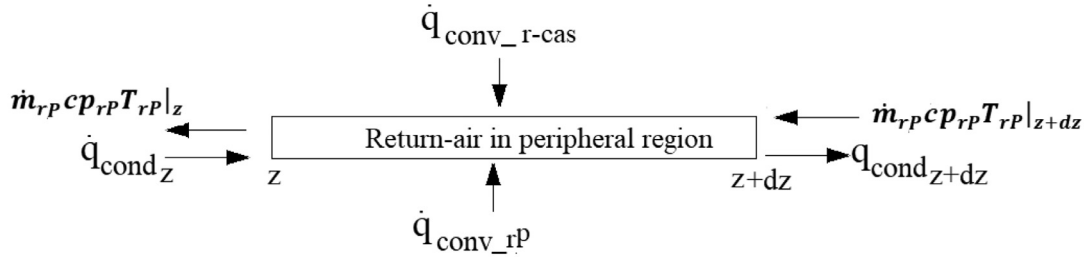


Fig. 12. A schematic showing heat transfer for the return air flowing through the peripheral region.

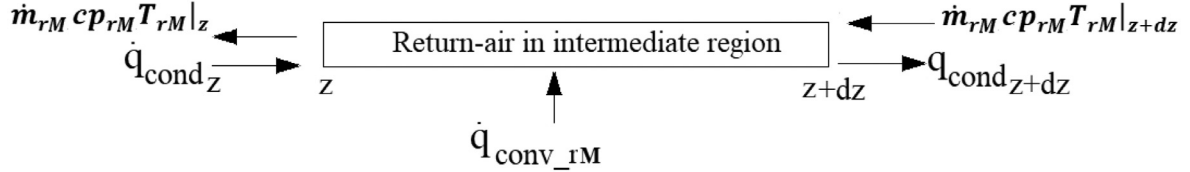


Fig. 13. A schematic showing heat transfer for the return air flowing through the intermediate region.

three regions, the mass flow rate should satisfy the following:

$$\frac{\dot{m}_{rp}}{A_{rp}} = \frac{\dot{m}_{rm}}{A_{rm}} = \frac{\dot{m}_{rc}}{A_{rc}}. \quad (12)$$

The Reynolds number (Re_{dh}), based on the hydraulic diameter of central, intermediate, and peripheral regions, will be different. Therefore, unlike the primary air, the heat transfer modeling for return air will be different for the different regions. Hence, the governing equations are derived for the same. These are described subsequently.

2.2.4.1. Return airflow through the peripheral region. The different modes of heat exchange, for return air flowing through the peripheral region, are depicted in Fig. 12. Here, $\dot{q}_{conv_rp}$ denotes the rate of convective heat transfer with the peripheral absorbers, $\dot{q}_{conv_r-cas}$ denotes the convective heat transfer with the casing, and $\dot{q}_{cond_z}$ denotes the rate of heat conduction at z for the adopted coordinate system. The rate of heat transfer by advection at z is denoted by $\dot{m}_{rp} c p_{rp} T_{rp}|_z$. It may be noted that the return airflow along the z -direction and is depicted by an arrow.

The instantaneous energy balance gives,

$$\underbrace{\frac{k_{rp}}{\rho_{rp} c p_{rp}} \frac{\partial^2 T_{rp}}{\partial z^2}}_{\text{conduction term}} + \underbrace{\frac{\dot{m}_{rp}}{\rho_{rp} A_{rp}} \frac{\partial T_{rp}}{\partial z}}_{\text{advection term in return air}} + \underbrace{\frac{4\pi h_{rp} r_{ab}}{\rho_{rp} c p_{rp} A_{rp}} (T_p - T_{rp})}_{\text{convection term from P to return-air}} + \underbrace{\frac{2\pi h_{rcas} r_{cas,i}}{3\rho_{rp} c p_{rp} A_{rp}} (T_{cas} - T_{rp})}_{\text{convection term from cas to return-air}} = \underbrace{\frac{\partial T_{rp}}{\partial t}}_{\text{unsteady term}} \quad (13)$$

The required initial and boundary conditions for solving the Eq. (13) are as follows:

Initial condition: The return air temperature in the periphery region is the same as ambient i.e. at $t = 0$, $T_{rp} = T_{atm}$

Boundary conditions: The defined boundary conditions at the return air inlet $z = L$ is $T_{rp} = T_{atm}$ and at the return air outlet $z = 0$ is $\frac{\partial T_{rp}}{\partial z} = 0$.

2.2.4.2. Return airflow through the intermediate region. The return air flowing through the intermediate region interacts with M and the modes of heat transfer are illustrated in Fig. 13. Here, $\dot{q}_{conv_rm}$ depicts the convective heat transfer from M to the return air. The other symbolic representations for conduction and advection are already described.

The instantaneous energy balance gives,

$$\underbrace{\frac{k_{rm}}{\rho_{rm} c p_{rm}} \frac{\partial^2 T_{rm}}{\partial z^2}}_{\text{conduction term}} + \underbrace{\frac{\dot{m}_{rm}}{\rho_{rm} A_{rm}} \frac{\partial T_{rm}}{\partial z}}_{\text{advection term in return air}} + \underbrace{\frac{2\pi h_{rm} r_{ab}}{\rho_{rm} c p_{rm} A_{rm}} (T_M - T_{rm})}_{\text{convection term from M to return-air}} = \underbrace{\frac{\partial T_{rm}}{\partial t}}_{\text{unsteady term}} \quad (14)$$

Initial condition: at $t = 0$ is $T_{rm} = T_{atm}$.

Boundary conditions: at $z = L$ is $T_{rm} = T_{atm}$ and at $z = 0$ is $\frac{\partial T_{rm}}{\partial z} = 0$.

2.2.4.3. Return airflow through the central region. The modes of heat transfer for the return air flowing through the central region are shown in Fig. 14. Here, $\dot{q}_{conv_rc}$ represents the convective heat transfer from the central absorber to the return air.

The instantaneous energy balance gives,

$$\underbrace{\frac{k_{rc}}{\rho_{rc} c p_{rc}} \frac{\partial^2 T_{rc}}{\partial z^2}}_{\text{conduction term}} + \underbrace{\frac{\dot{m}_{rc}}{\rho_{rc} A_{rc}} \frac{\partial T_{rc}}{\partial z}}_{\text{advection term in return air}} + \underbrace{\frac{\pi h_{rc} r_{ab}}{3\rho_{rc} c p_{rc} A_{rc}} (T_C - T_{rc})}_{\text{convection term from C to return-air}} = \underbrace{\frac{\partial T_{rc}}{\partial t}}_{\text{unsteady term}} \quad (15)$$

Initial condition: Initially the temperature of return air in the central region is equal to ambient temperature.

At $t=0$; $T_{rc} = T_{atm}$

Boundary condition: The temperature of return air in its inlet i.e. at $z=L$ is ambient. At the outlet of the central region i.e. $z=0$, the adiabatic boundary condition is considered.

At $z=L$; $T_{rc} = T_{atm}$

At $z=0$; $\frac{\partial T_{rc}}{\partial z} = 0$

The required heat transfer coefficient for return air flowing around the P, M, and C are computed using $Re_{dh_r} = \frac{dh_r \dot{m}_r}{A_r \mu}$ [54] and is as follows:

$$Nu_r = 0.345 * Re_{dh_r}^{0.597}. \quad (16)$$

Here, the convective heat transfer coefficient $h_r = Nu_r(k_r/dh_r)$ for the return air.

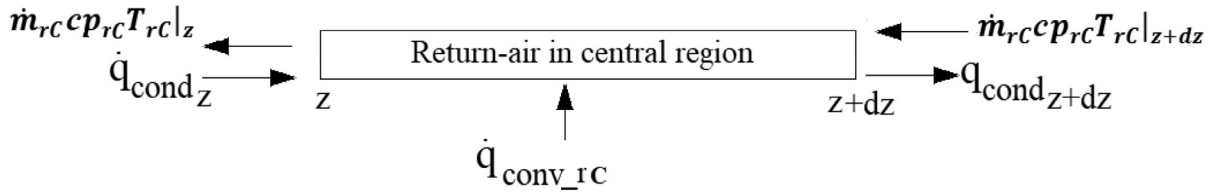


Fig. 14. A schematic showing heat transfer for the return air flowing through the central region.

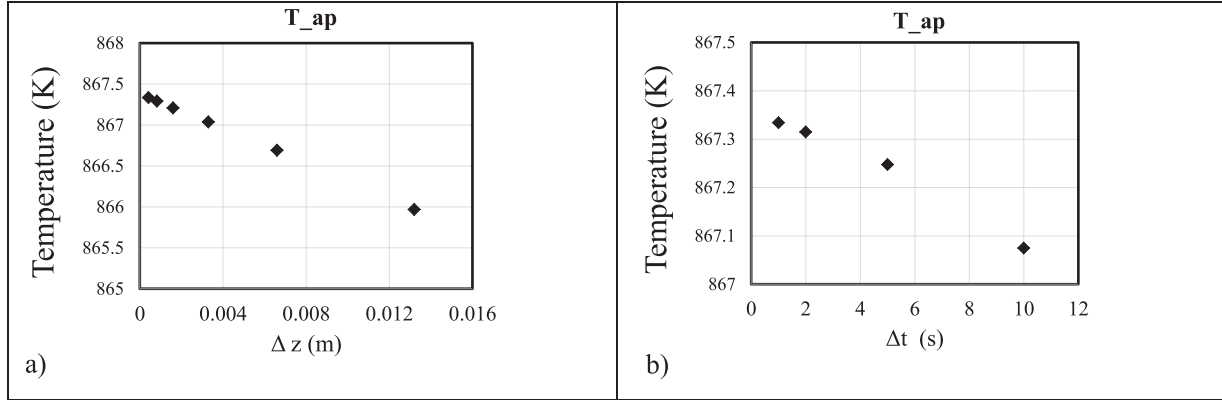


Fig. 15. The computed primary air temperature at the outlet of P at 5000 seconds for different a) grid sizes and b) time steps.

2.3. Grid Independence test and validation

A laboratory experiment for an OVAR design [7,8], with one central absorber C and six peripheral absorbers P having 52% porosity, is selected for the grid independence test and validation of the developed 1D multi-zone model. The experimental conditions are input power $\dot{q} = 1500$ W, air mass flow rate $\dot{m}_a = 0.005$ kg/s, and the air return ratio $ARR = 0.6$ [44] obtained using the detailed air temperature measurement. Calculations are performed with different grid sizes (Δz) and time steps (Δt), and the obtained values at 5000 seconds are shown in Fig. 15. The computed values for the primary air temperature at the outlet of P (T_{ap}) at 5000 seconds, for $0.0004 \text{ m} \leq \Delta z \leq 0.0016 \text{ m}$ and $\Delta t = 1 \text{ s}$, are $867.20 \text{ K} \leq T_{ap} \leq 867.33 \text{ K}$ and the percentage relative variation is $\leq 0.02 \%$. Thus, $\Delta z = 0.0004 \text{ m}$ is selected for the time-independence test. Subsequently, the different time-steps ($1 \text{ s} \leq \Delta t \leq 10 \text{ s}$) are considered for $\Delta z = 0.0004 \text{ m}$ and the computed values for T_{ap} at 5000 seconds are $867.07 \text{ K} \leq T_{ap} \leq 867.33 \text{ K}$ and the relative percentage variation $\leq 0.019 \%$ for $\Delta t = 1 \text{ s}$. Therefore, a time-step $\Delta t = 1 \text{ s}$ is selected for further simulations.

Fig. 16a shows a comparison between the computed and the measured primary air temperature at the outlet of C and P. The measured, time-dependent, inlet air temperature is used as a boundary condition for primary air. The comparison also reveals that the transient heating of primary air up to 300 seconds and the subsequent steady-state up to 2400 seconds are reproduced fairly well. Furthermore, the average values of measured primary air temperature at the outlet for all the 7 absorbers, at a time interval of 250 seconds, are compared with the computed values in Fig. 16b. This shows that the deviations between the measured and computed average values are well within the experimental uncertainty of about $\pm 7\%$. The highest and lowest deviations are 14°C at 750 seconds and 7°C at 250 seconds, respectively. Further analyses for a smaller time interval revealed that the computed and experimental data are within the same uncertainty range. Moreover, Fig. 16c revealed that the computed, steady-state, axially varying, primary air temperature for C is consistent with the reported two-

zone model with one C and six P [44]. The multi-zone model is now applied for the design assessment and scale-up of an OVAR design [7,8] with several intermediate absorber layers. Here, the estimated $ARR = 0.6$ for the selected OVAR design, from the laboratory experiments, is adopted. Furthermore, a parametric investigation is performed for a wide-range of ARR (0.3–0.8) to determine its effect on the performance of scale-up OVAR. These are presented subsequently.

3. Results and Discussions

This section reports the assessment of scale-up OVAR designs with one, two, and three intermediate absorber layers (Fig. 3) for a given total air mass flow rate $\dot{m}_a = 0.966$ kg/s through OVAR and input power $\dot{q} = 0.58$ MW or $\frac{\dot{q}}{\dot{m}_a} = 600$ kJ/kg. The selection of these parameters is motivated by the heat treatment inside a retrofitted solar convective furnace [55]. An absorber with diameter $d_{ab} = 0.3292 \text{ m}$, length $L = 0.3292 \text{ m}$ and porosity $\varepsilon = 0.52$ is selected based on the preliminary design analyses with one, two, or three intermediate absorber layers. In the analyses, the Reynolds number based on pore diameter ($Re_{d_{pr}}$) is maintained. These details are summarized in Table 2, which depicts that the number of straight pores increases substantially from 242 for the three intermediate absorber layers to 2500 for the one intermediate absorber layer. In other words, having several intermediate absorber layers seems more realistic than a single layer for fabrication. However, this will enhance the exposed front surface area of the absorbers to ambient and thus the heat losses. Therefore, a careful assessment of the OVAR designs is necessary for optimization. Considering such aspects, this section reports the effect of a) number of intermediate absorber layers, b) gap between absorbers, c) absorber length, d) volumetric and non-volumetric heating, e) absorber porosity and f) air return ratio on the thermal efficiency of OVAR.

Assuming that the entrance length is much smaller than the absorber pore length, the pressure drop across a straight, circular, pore is estimated using the following equation for laminar flow at

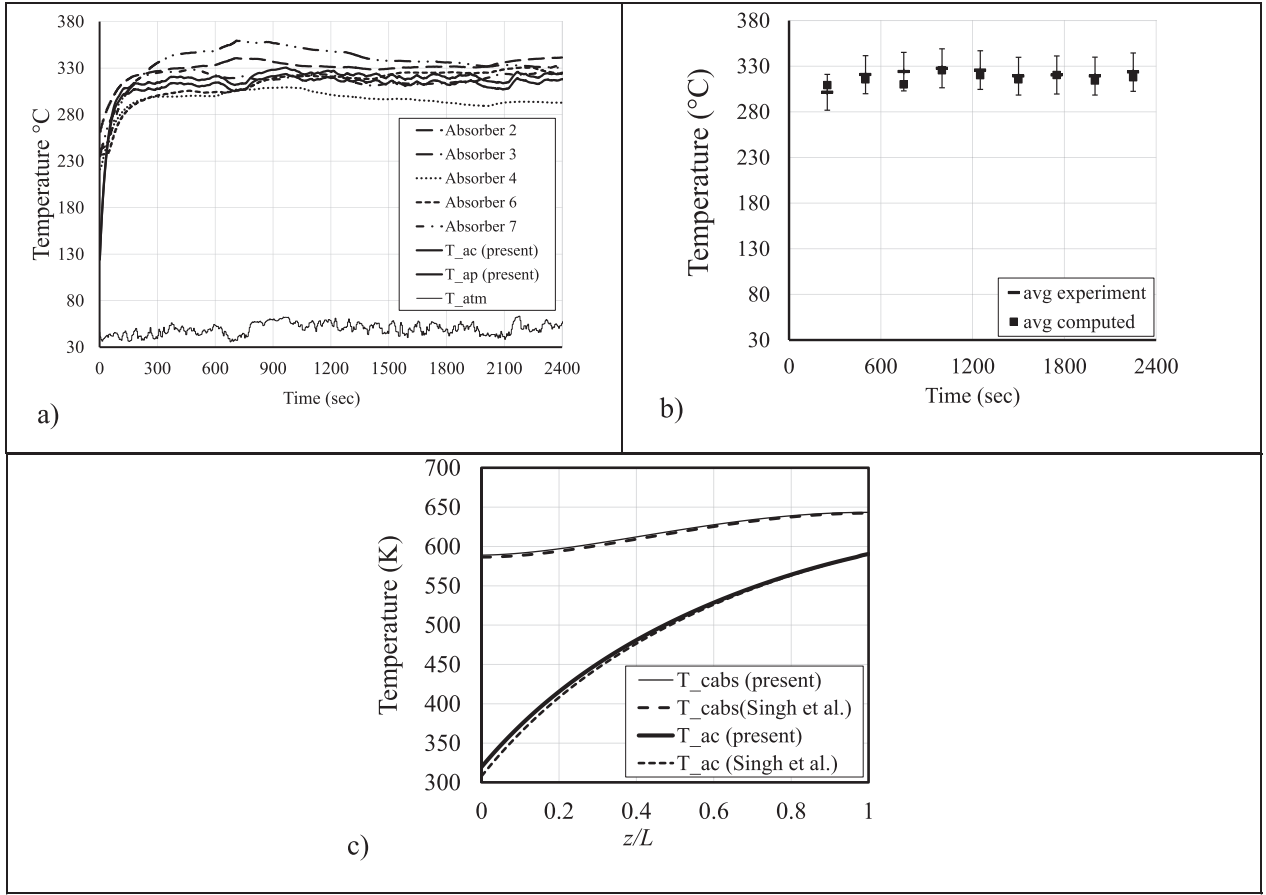


Fig. 16. A comparison between the computed and the measured [7,8] (a) primary air temperature at the outlet for C and P, b) average primary air temperature values for all the absorbers at a time interval of 250 seconds, c) comparison of computed temperature between present and two-zone model as in Singh et al. [43].

Table 2

The selected absorber parameters, to maintain, $Re_{d_{pr}}$, for the different numbers of intermediate absorber layers.

Number of Intermediate absorber layers	$\dot{m}_a$ (kg/s)	N_{ab}	d_{ab} (m)	D_{cas} (m)	$\dot{m}_{abs} = \dot{m}_a/N_{ab}$ (kg/s)	np	d_{pr} (m)	$Re_{d_{pr}}$
One	0.966	19	0.3292	2.536	0.0508	2500	0.0047	86
Two	0.966	37	0.3292	3.458	0.0261	660	0.0091	86
Three	0.966	61	0.3292	4.380	0.0158	242	0.0151	86

a small $Re_{d_{pr}} = 86$ [56]:

$$\Delta p_{total} = \underbrace{K_{in} \frac{Q_{pr}^2}{2gA^2}}_{\text{sudden contraction}} + \underbrace{\frac{128\mu L Q_{pr}}{d_{pr}^4}}_{\text{for a straight pore}} + \underbrace{K_{out} \frac{Q_{pr}^2}{2gA^2}}_{\text{sudden expansion}} \quad (17)$$

Here, Δp_{total} is the total pressure drop across a straight pore, Q_{pr} is the volume flow rate in a single pore, K_{in} and K_{out} are the loss coefficients, based on entrance and exit diameter ratio, and are equal to 0.24 [56]. The calculated values in Fig. 17 depict the highest pressure drop for the absorber design with 3 intermediate layers and the lowest pressure drop for the absorber design with 1 intermediate layer. Generally, these pressure-drop values are much smaller than that of the complex porous form type absorbers [10], and decrease exponentially for d_{ab}/L up to 1 and then rather slowly. Therefore, selecting $d_{ab}/L \geq 1$ will be preferred and pre-assessment of pressure-drop is recommended for the continuous operation of OVAR.

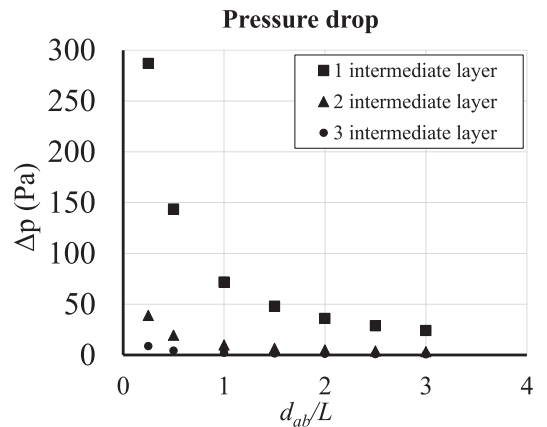


Fig. 17. Pressure drop in pore with respect to d_{ab}/L .

3.1. Volumetric heating

The ambient air mixed with return air enters the absorber pores as primary air and gains the heat from volumetrically heated

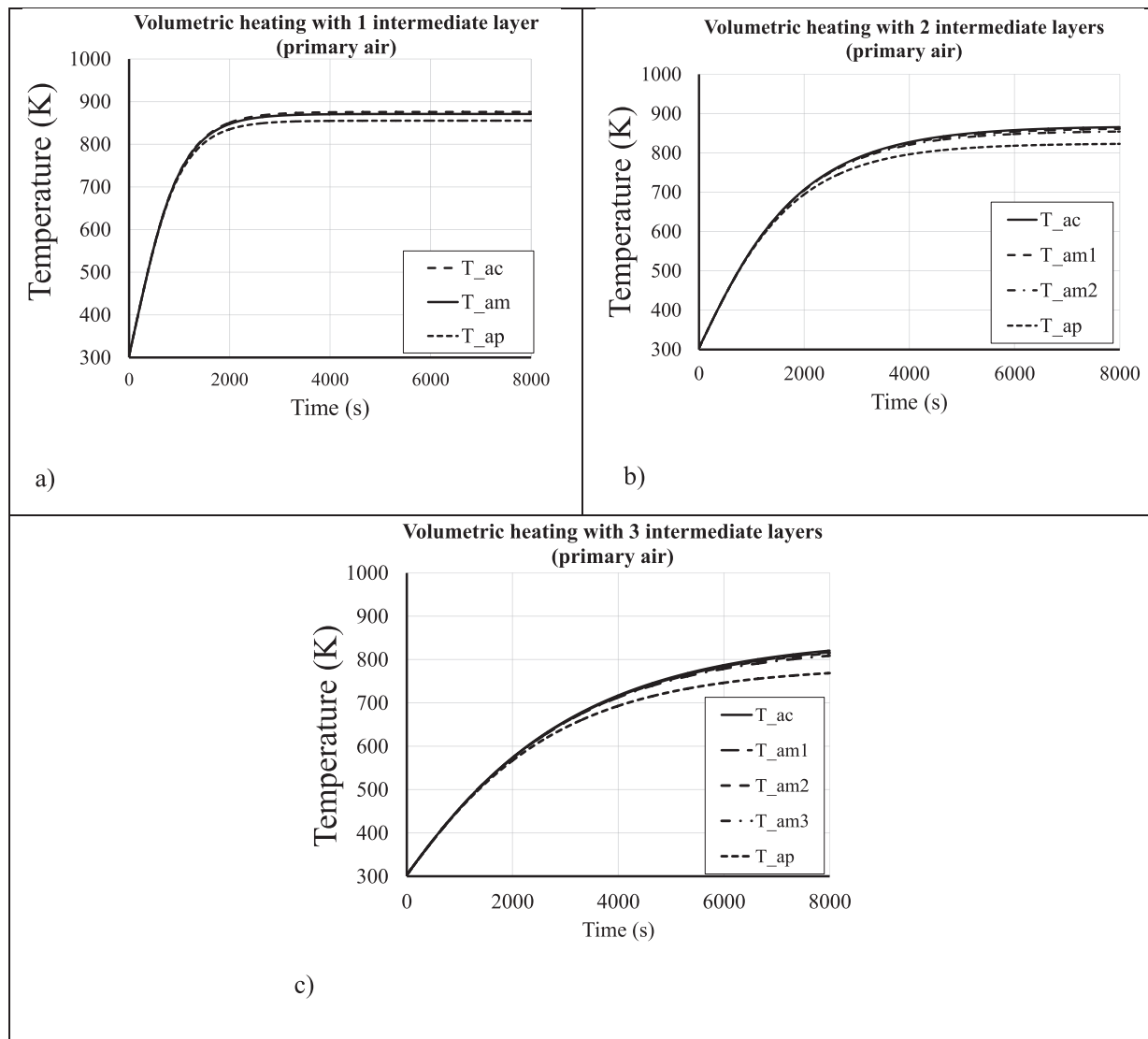


Fig. 18. Time-dependent primary air temperature at the outlet of C, M and P for a) one intermediate absorber layer, b) two intermediate absorber layers and c) three intermediate absorber layers for volumetric heating.

absorbers, and leaves through the absorber outlets. The time-dependent, average, primary air temperature at the outlet of C, M, and P is shown in Fig. 18. In the provided legends, T_a stands for air temperature and the second subscript, c, m, and p, stands for central, intermediate, and peripheral absorbers. Following may be inferred:

- for one intermediate absorber layer, the air temperature increases rapidly from 300 K to 850 K up to 2000 seconds and gradually increases up to 870 - 875 K until 4000 seconds at the steady-state.
- for two intermediate absorber layers (m1 and m2), the air temperature rapidly increases, from 300 K to 830 K for C and M and from 300 K to 800 K for P, up to 4000 seconds. Therefore, the rate of temperature rise decreases with the increasing number of intermediate absorber layers. This is attributed to the elevated heat loss from the exposed, front, surface area of absorbers. Subsequently, the primary air temperature attains 840-875 K at the steady-state around 6000 seconds.
- for three intermediate absorber layers (m1, m2, and m3), the air temperature gradually increases from 300 K to 830 K until 8000 seconds. This is consistent with the findings from one and two intermediate absorber layers-based simulations. Hence,

it may be concluded that the use of several intermediate absorber layers (M) benefits from a lower surface temperature, however, for a higher heat loss. This aspect will be investigated at a greater depth at a later stage.

The axial variations for absorbers and primary air temperatures, from the inlet ($z/L = 0$) to the outlet ($z/L = 1$) for the absorbers, are provided in Fig. 19 at 8000 seconds. As expected for volumetric heating, the absorber temperature increases from inlet to outlet, and the primary air temperature follows the same trend. In the legend, T_C, T_M, and T_P stand for the absorber temperature for C, M, and P, respectively. The primary air temperatures at the outlet for C, M, and P are identified with T_{ac}, T_{am} and T_{ap}, respectively. The following may be inferred from this figure:

- for one intermediate absorber layer, the absorber temperature increases from 460 K at $z/L = 0$ to 890 K at $z/L = 1$. The air temperature increases from 310 K at $z/L = 0$ to 870 K at $z/L = 1$.
- for two intermediate absorber layers, the absorber temperature for C and M increases from 580 K at $z/L = 0$ to 890 K at $z/L = 1$, and that of P increases from 565 K to 850 K. The air temperature for C and M increases from 317 K at $z/L = 0$ to 861 K

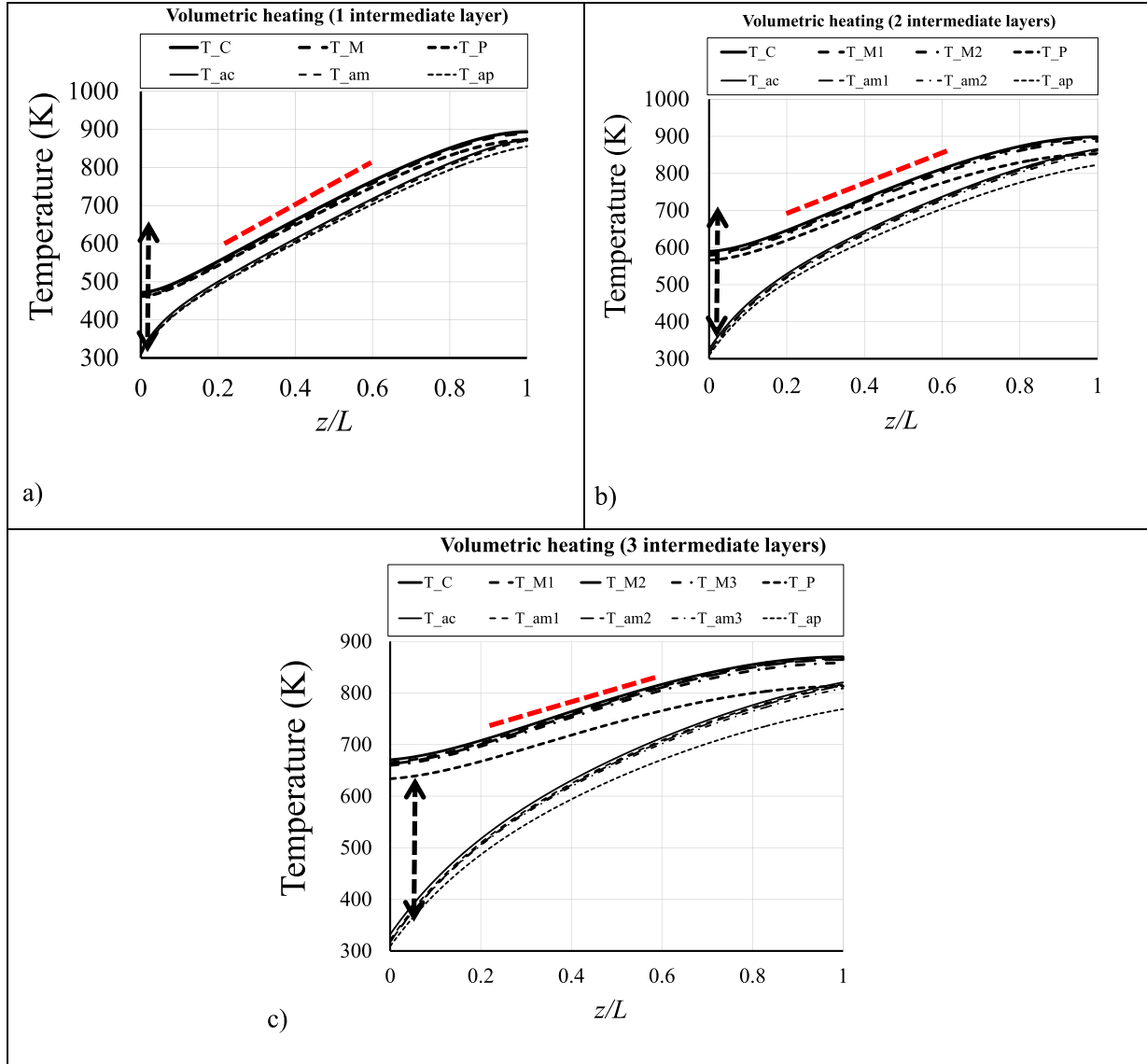


Fig. 19. Axial variation of primary air and absorber temperature at 8000 seconds for a) one intermediate absorber layer, b) two intermediate absorber layers and c) three intermediate absorber layers for volumetric heating.

- at $z/L = 1$. The air temperature for P increases from 310 K at $z/L = 1$ to 823 K at $z/L = 1$.
- c. for three intermediate absorber layers, the absorber temperature for C and M increases from 664 K at $z/L = 0$ to 865 K at $z/L = 1$ and that of P increases from 633 K at $z/L = 0$ to 812 K at $z/L = 1$. The air temperature for C and M increases from 321 K at $z/L = 1$ to 815 K at $z/L = 1$ and for P from 309 K at $z/L = 1$ to 768 K at $z/L = 1$.
- d. the difference between the absorber and primary air temperature increases with the number of layers, represented by a vertical dotted double-headed arrow, for the same total air mass flow rate $\dot{m}_a = 0.966$ kg/s. This is attributed to the calculated, surface-area averaged, convective heat transfer coefficients for the internal flow in a pore ($h_{d,pr}$) 48.5 W/m²K, 25.3 W/m²K, and 15.6 W/m²K for one, two and three intermediate absorber layers, respectively.
- e. the absorber temperature gradient along the axial direction (indicated by inclined dotted line) is the least for three intermediate absorber layers. This is expected to reduce the generated thermal stress and enhance the durability of OVAR for several intermediate absorber layers. Also, the multiple intermediate

absorber layers may be selected considering fabrication difficulties with a large number of pores. This means a balance between the losses and durability is the key for design optimization. The heat losses for the OVAR designs with 1, 2, and three intermediate absorber layers are performed subsequently.

The maximum radiation-based heat loss from the exposed front surface of the absorbers is estimated using the following equation:

$$\dot{q}_{rad} = \epsilon_{ab} \sigma A_{front} (T_{ab}^4 - T_{atm}^4) \quad (18)$$

where ϵ_{ab} is the emissivity of absorber material, A_{front} is the exposed aperture of OVAR, σ is the Stefan-Boltzmann constant, and T_{ab} is the absorber surface temperature at $z=0$. The ϵ_{ab} may be expressed as a function of temperature and porosity, if available, for a particular absorber geometry. In the presented simulations $\epsilon_{ab} = 0.6$ is considered for a conservative assessment of the receiver efficiency. The natural convection based heat loss from the exposed absorber surface is obtained using the following equation:

$$\dot{q}_{nat} = h_{nat} A_{front} (T_{abs} - T_{atm}) \quad (19)$$

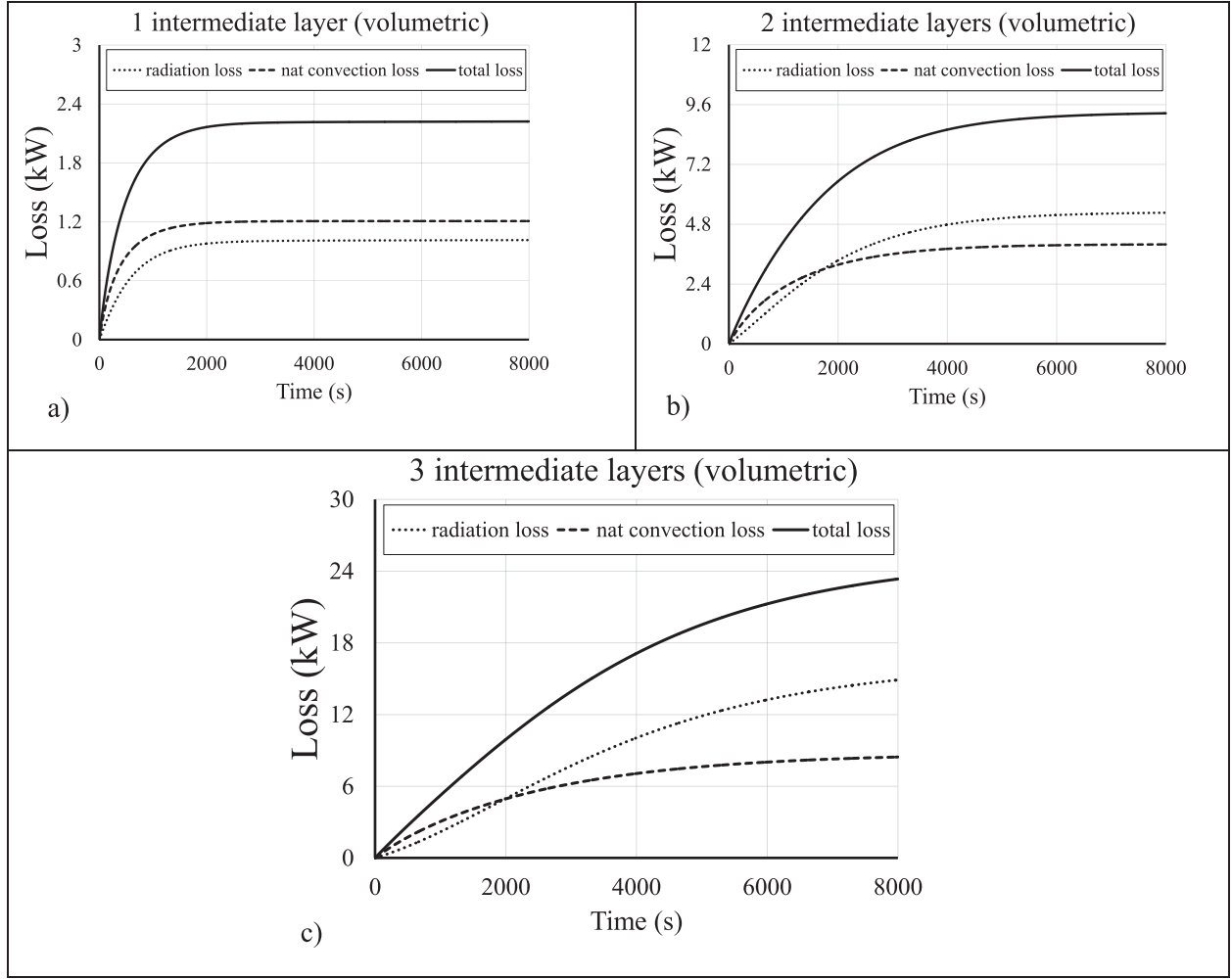


Fig. 20. Heat losses from OVAR for volumetric heating with a) one intermediate absorber layer, b) two intermediate absorber layers, and c) three intermediate absorber layers.

where the heat transfer coefficient (h_{nat}) is obtained using the following Nusselt number correlation [53]:

$$Nu_{nat} = \left[0.60 + \frac{0.378Ra_d^{1/6}}{\{1 + (0.559/Pr)^{9/16}\}^{8/27}} \right]^2 \quad (20)$$

The computed heat losses for volumetric heating of the absorbers with one, two and three intermediate absorber layers are shown in Fig. 20. Following may be observed:

- for one intermediate absorber layer, the radiation and natural convection-based heat losses are comparable.
- for 2 and three intermediate absorber layers, the radiation-based heat loss is 1-2 times that of the natural convection at around 8000 seconds. This means radiation-based heat loss is likely to dominate in an OVAR with several intermediate absorber layers.
- generally, the increasing trend of heat losses is attributed to the rising absorber temperature during the unsteady phase.

3.2. Non-volumetric heating

The computed, time-dependent, primary air temperatures at the outlet of C, M and P are shown in Fig. 21. The air temperature difference between C, M, and P for non-volumetric heating is higher than volumetric heating. In other words, thermal

non-uniformity will be enhanced for non-volumetric heating. Thus, efforts are ongoing to homogenize the concentrated beam radiation with sophisticated optical devices like homogenizers, onto the OVAR aperture. The approximation to estimate the extinction coefficient (β) is adopted from [49]. Here, the parameter c for $\beta = \frac{c(1-\varepsilon)}{d_{pr}}$ is reported as variable (see e.g. [57,58]). Considering these findings $2 \leq c \leq 5$ is numerically varied to calculate the thermal efficiency (η_{th}) for the case with one intermediate absorber layer. The calculated thermal efficiencies are $0.74 \leq \eta_{th} \leq 0.76$ for non-volumetric heating conditions. The change in computed thermal efficiency of about 0.02 is quite small and is consistent with the reported findings of [58]. Therefore, $c = 3.0$ is maintained for simulations with this known uncertainty. The heat flux on the solid absorber surface is 507 – 640 Suns for 1 intermediate layer, 513–627 Suns for 2 intermediate layers and 528–612 Suns for 3 intermediate layers for the non-volumetric heating condition. Following may be inferred from Fig. 21a, 21b, and 21c:

- for one intermediate absorber layer, the air temperature rapidly increases from 300 K to 685 K for P, 300 K to 750 K for M and 300 K to 790 K for C up to 2000 seconds. At the steady-state, the temperature difference between C and P exceeds 110 K, and the maximum air temperature is higher than 800 K.
- for two intermediate absorber layers (denoted by m1 and m2), the air temperatures increase from 300 K to 640 K for P, 300 K to 700/723 K for m1/m2 and 300 K – 730 K for C. At the steady-

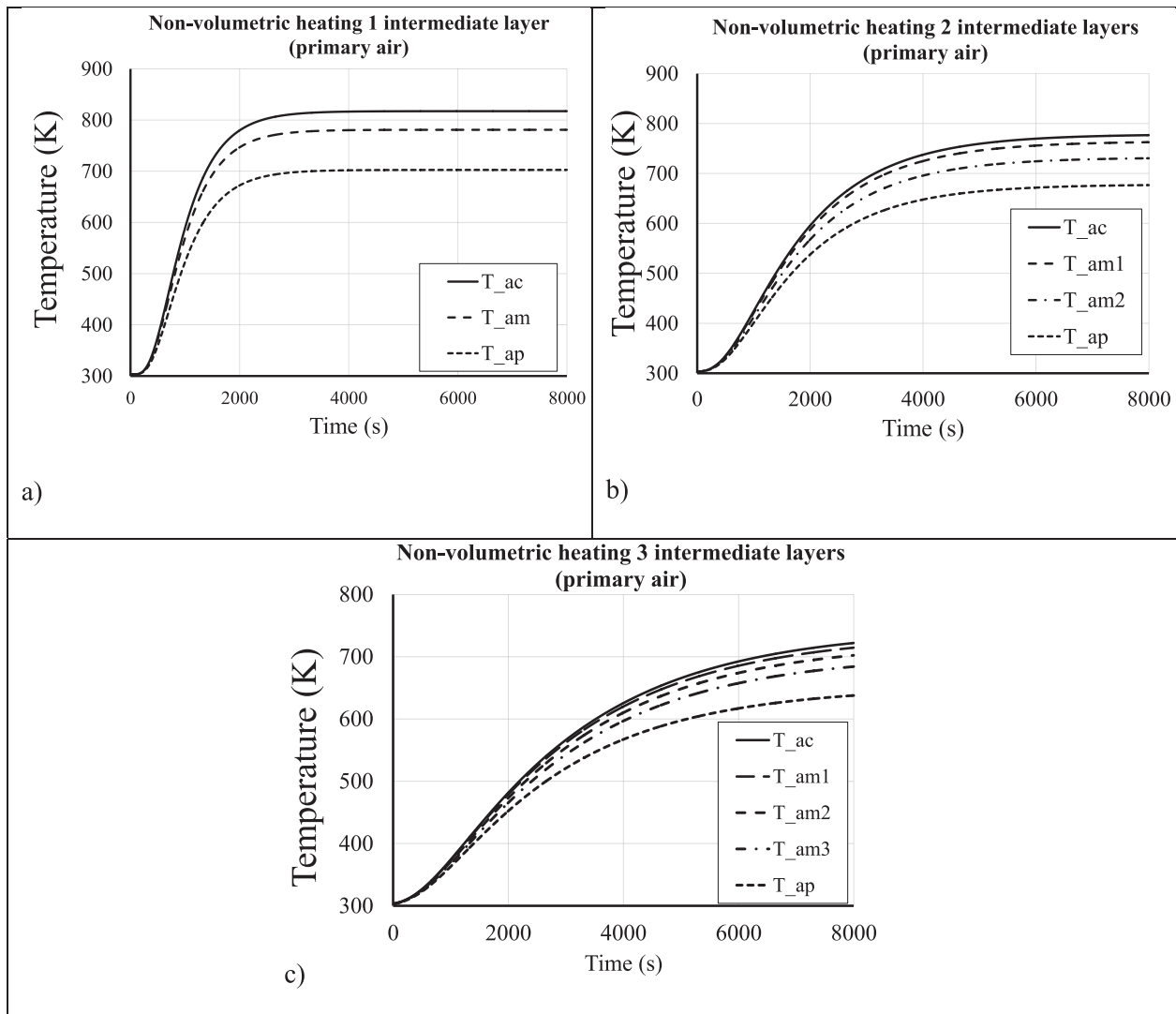


Fig. 21. Temporal variation of primary air temperature at the absorber outlet for a) one intermediate absorber layer, b) two intermediate absorber layers and c) three intermediate absorber layers for non-volumetric heating.

state, the maximum temperature difference between C and P is around 95 K, and the maximum air temperature is around 780 K. This means, with the increasing number of intermediate absorber layers (i) the maximum air temperature decreases and (ii) the thermal non-uniformity is also mitigated. These are attributed to the redistribution of heat flux onto the absorbers arranged in several layers (see Fig. 2).

- c. for three intermediate absorber layers, the maximum air temperature at around 8000 seconds is around 730 K, and the maximum temperature difference between C and P is around 80 K. This reconfirms the effect of multiple intermediate absorbers layers on the heating up of air.

The computed absorber and primary air temperatures at 8000 seconds, along the axial direction, are shown in Fig. 22 for OVAR design with one, two, and three intermediate absorber layers. As expected, the non-volumetric heating results in the highest temperature at the inlet $z/L = 0$ and the lowest temperature at the outlet $z/L = 1$. The primary air temperature increases from $z/L = 0$ to $z/L = 1$ on account of the heat transfer from the heated absorbers. The following may be inferred from this figure:

- a. the thermal development length for air increases with the number of intermediate absorber layers. Consequently, the pri-

mary air attains the absorber temperature at around $z/L = 0.25$ for one intermediate absorber layer and $z/L = 0.65$ for three intermediate absorber layers. An important observation is that the absorber length is to be carefully estimated depending on the number of intermediate absorber layers.

- b. the maximum absorber temperature at the inlet ($z/L = 0$) reduces with the increasing number of intermediate absorber layers. As a consequence, the primary air attains the lowest temperature for the three intermediate absorber layers.
- c. the absorber temperature gradient, along the axial direction, is the highest for one intermediate absorber layer. Therefore, for an OVAR design, the thermal stresses will increase with the decreasing number of the intermediate absorber layers. As a consequence, the use of multiple intermediate absorber layers will be beneficial for the durability of an OVAR design.
- One important question arises, whether the heat loss will be lower for a higher number of intermediate absorber layers? This aspect will be investigated subsequently.

The computed, time-dependent, heat losses via radiation and natural convection are shown in Fig. 23. The increasing heat loss with the number of intermediate absorber layers is attributed to the increasing exposed surface area of absorbers to atmospheric

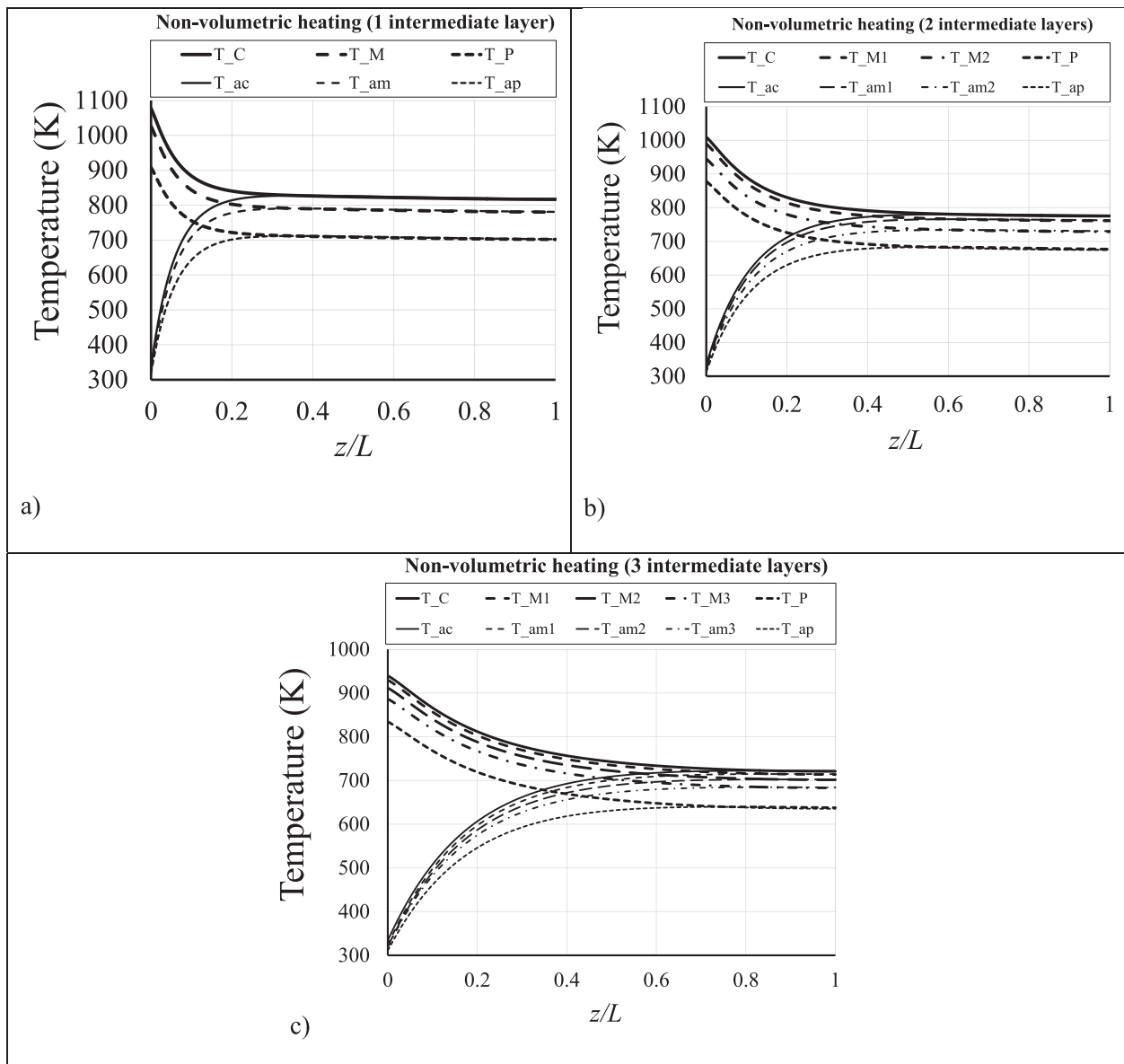


Fig. 22. Axial variation of primary air and absorber temperature at 8000 seconds, for OVAR design with a) one intermediate absorber layer, b) two intermediate absorber layers and c) three intermediate absorber layers, for non-volumetric heating.

conditions. Also, the heat loss for non-volumetric heating is substantially greater than that of volumetric heating on account of the exposed surface temperature. An important observation is that the radiation-based heat loss is 3–4 times that of the natural convection. This means an appropriate coating with a low emissivity for thermal radiation and a high absorptivity for solar radiation should be employed, especially at the exposed surface of absorbers, for non-volumetric heating. Therefore, further improvement of the developed multi-zone model, including thermo-optical coupling, is foreseen.

3.3. Parametric investigation for an OVAR design

So far, detailed investigations are performed to assess the effect of volumetric and non-volumetric heating on the absorber temperature, primary air temperature, and heat losses. In this sub-section, investigations are performed to assess the effect of important parameters, such as air return ratio (ARR), power to air mass flow rate ratio ($\dot{q}/\dot{m}_a$), absorber diameter to length ratio, absorber gap

to length ratio, porosity, and return air temperature on the thermal efficiency of an OVAR, for volumetric and non-volumetric heating conditions. This is a step towards the scale-up of an OVAR, design optimization, and recommending the operating conditions for high thermal efficiency and durability.

3.3.1. Effect of Air Return Ratio

Air return ratio (ARR) is defined as the fraction of return air, mixed with atmospheric air, which is sucked into the absorber pores as the primary air. Fig. 24 shows that the thermal efficiency of an OVAR increases with the increasing ARR for volumetric and non-volumetric heating, inlet return air temperature $T_{r,in} = 300$ K, $d_{ab}/L = 1$ and $G/L = 0.4$. Therefore, ideally, a very high ARR should be preferred, which will promote internal cooling and thermal uniformity for absorbers. However, a high ARR will reduce the temperature of primary air, a mixture of ambient air and return air, at the absorber inlet. Consequently, a marginal reduction in the outlet primary air temperature is observed, for $ARR > 0.6$, in OVAR with three intermediate absorber layers. Thus an important finding

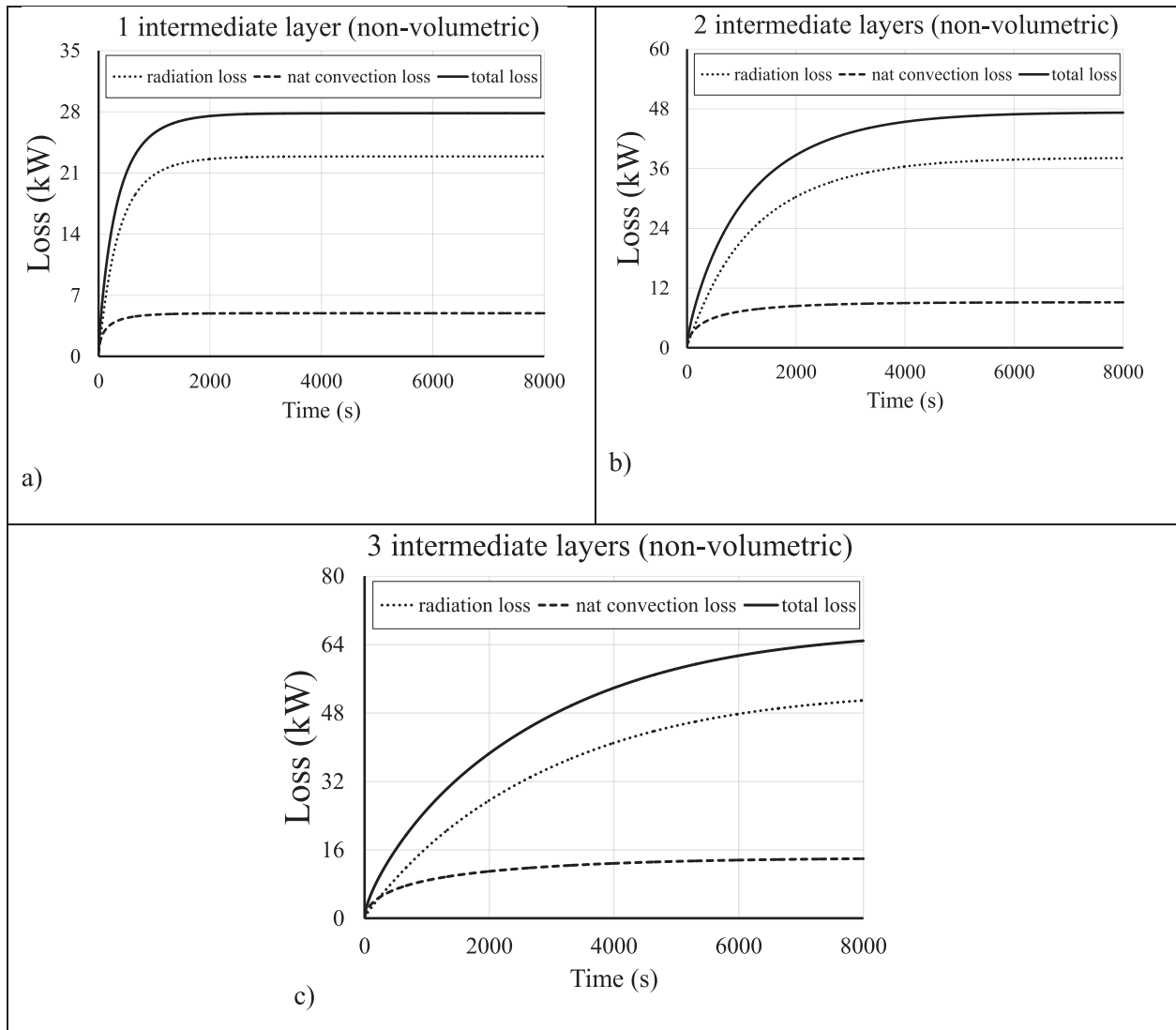


Fig. 23. The heat losses from OVAR for non-volumetric heating with a) one intermediate absorber layer, b) two intermediate absorber layers and c) three intermediate absorber layers.

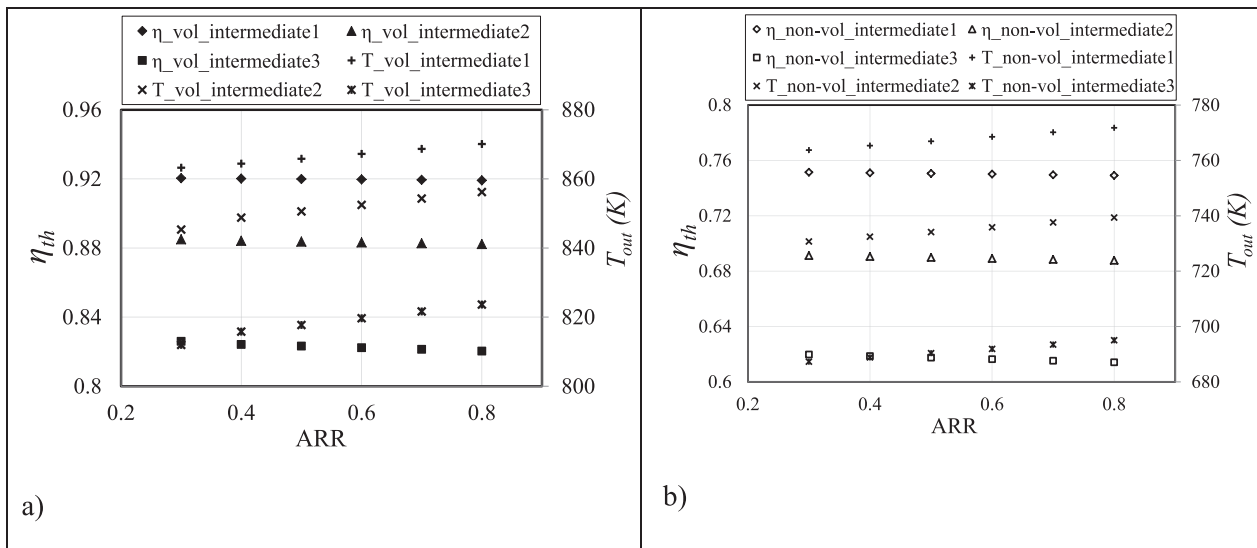


Fig. 24. The effect of air return ratio (ARR) on the OVAR thermal efficiency and the outlet temperature of primary air with one, two and three intermediate absorber layers, for (a) volumetric heating and (b) non-volumetric heating with $\frac{q}{m_a} = 600$ kJ/kg.

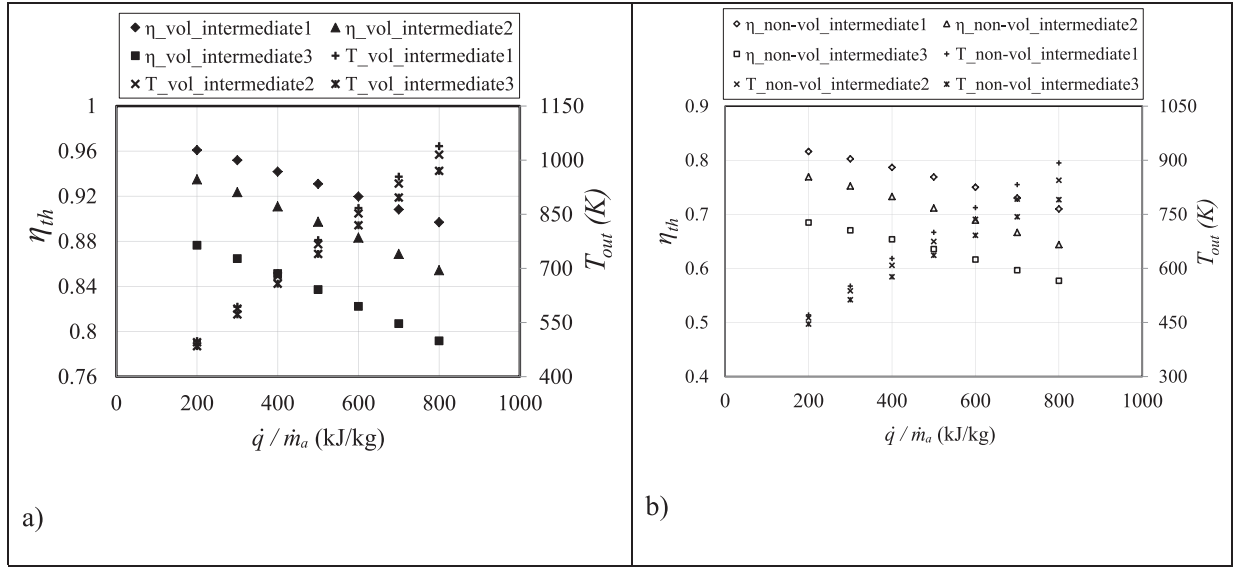


Fig. 25. The effect of $\dot{q}/\dot{m}_a$ on the OVAR thermal efficiency (η_{th}) and the outlet temperature of primary air (T_{out}) with one, two and three intermediate absorber layers, for (a) volumetric heating and (b) non-volumetric heating.

is that the ARR should be carefully adopted for an OVAR design, pumping power, and outlet temperature requirement. As a compromise, an ARR = 0.6 is recommended for the prescribed OVAR design with three intermediate absorber layers.

3.3.2. Effect of $\dot{q}/\dot{m}_a$

The computed thermal efficiency (η_{th}) and primary air outlet temperature (T_{out}), for $200 \leq \frac{\dot{q}}{\dot{m}_a} \leq 800$ ($\frac{\text{kJ}}{\text{kg}}$), different heating conditions, $T_{r,in} = 300$ K, $d_{ab}/L = 1$ and $G/L = 0.4$ and ARR = 0.6 are shown in Fig. 25. Generally, the thermal efficiency decreases with increasing $\frac{\dot{q}}{\dot{m}_a}$ for volumetric and non-volumetric heating conditions. Following may be inferred:

- for volumetric heating, $0.9 \leq \eta_{th} \leq 0.96$ for one intermediate absorber layer and $0.79 \leq \eta_{th} \leq 0.88$ for three intermediate absorber layers. For non-volumetric heating, $0.71 \leq \eta_{th} \leq 0.81$ for one intermediate absorber layer and $0.58 \leq \eta_{th} \leq 0.69$ for three intermediate absorber layers. The thermal efficiency for an OVAR design decreases with the increasing number of intermediate absorber layers for volumetric and non-volumetric heating.
- the outlet temperature of primary air is higher for volumetric heating, in comparison to that of the non-volumetric heating, in an OVAR. Therefore, an OVAR with non-volumetric heating should be operated at a much higher $\frac{\dot{q}}{\dot{m}_a}$ in comparison to the volumetric heating to attain the desired primary air temperature.

3.3.3. Effect of absorber diameter to length ratio

Fig. 26 shows the variation of OVAR thermal efficiency (η_{th}) and the outlet temperature of primary air (T_{out}) for different absorber diameter to length ratios ($0.5 \leq d_{ab}/L \leq 3$), different heating conditions, $T_{r,in} = 300$ K, $G/L = 0.4$, and ARR = 0.6. The following may be inferred from this figure:

- η_{th} increases for $0.5 \leq d_{ab}/L \leq 1$ with volumetric heating and $0.5 \leq d_{ab}/L \leq 2$ with non-volumetric heating for two and three intermediate absorber layers. Therefore, a relatively smaller absorber diameter should be preferred for volumetric heating in comparison to non-volumetric heating conditions. A sharp rise in η_{th} for $0.5 \leq d_{ab}/L \leq 1$ is an indication that the absorber length is inadequate to attain the highest air temper-

ature, which leads to insufficient heat recovery. Also, the decay in η_{th} for $2 \leq d_{ab}/L \leq 3$ indicates that the absorber length is more than adequate, and the absorber surface area is substantial for heat loss.

- apparently, for both the heating conditions, $1 \leq d_{ab}/L \leq 2$ may be recommended for the scale-up OVAR design with three intermediate absorber layers. Similar investigations are recommended for selecting an appropriate absorber length for other OVAR designs.

3.3.4. Effect of absorber gap to length ratio

Fig. 27 depicts the variation of η_{th} and T_{out} for different absorber-gap to length ratios ($0.05 \leq G/L \leq 0.3$), $T_{r,in} = 300$ K, $d_{ab}/L = 1$, and ARR = 0.6. It may be noted that G/L determines the OVAR aperture and the return airflow regime around the absorbers. A small G/L implies densely packed, and a high G/L implies loosely packed absorbers in an OVAR. The increasing η_{th} for $0.05 \leq G/L \leq 0.2$, reveals that the relative heat recovery is higher than that of the heat losses from an OVAR aperture. It is worthwhile to note that η_{th} reaches a practical saturation at $G/L = 0.2$ for the OVAR designs with one, two and three intermediate absorber layers, and for different heating conditions. Therefore, $0.15 \leq G/L \leq 0.25$ may be recommended for the scale-up OVAR design with several intermediate absorber layers.

3.3.5. Effect of absorber porosity

Fig. 28 depicts the variation of η_{th} and T_{out} for different absorber porosities ($0.3 \leq \varepsilon \leq 0.8$), different heating conditions, $T_{r,in} = 300$ K, $d_{ab}/L = 1$, $G/L = 0.4$, and ARR = 0.6. The variations in η_{th} and T_{out} with porosity, for volumetric and non-volumetric heating effects, follow different trends. For volumetric heating, η_{th} increases rapidly for $0.3 \leq \varepsilon \leq 0.6$, and attains a practical limit. For non-volumetric heating η_{th} increases monotonically with ε , however, the outlet primary air temperature follows a different trend. Therefore, ε for an absorber with the straight pore and volumetric heating should be around 0.6, and ε for an absorber with non-volumetric heating should be as high as possible. However, with increasing ε , the material strength and therefore the durability of an absorber will be affected. Thus, prescribing an upper limit for ε will be dependent on the selected absorber material.

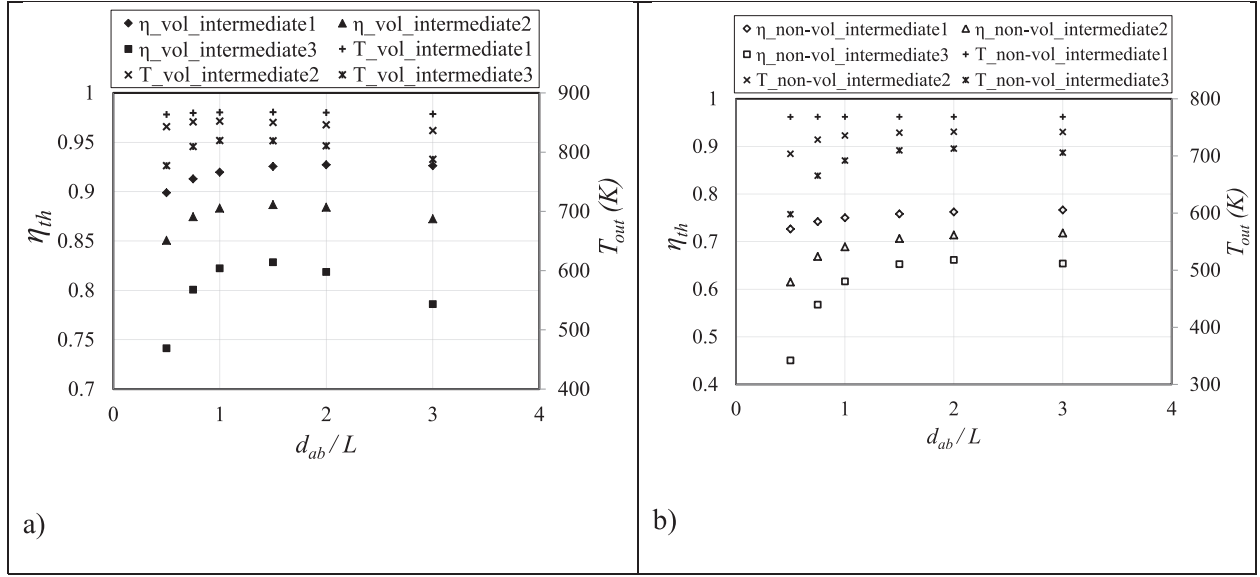


Fig. 26. The effect of d_{ab}/L on the OVAR thermal efficiency (η_{th}) and the outlet temperature of primary air (T_{out}) with one, two and three intermediate absorber layers, for (a) volumetric heating and (b) non-volumetric heating with $\frac{q}{m_a} = 600$ kJ/kg.

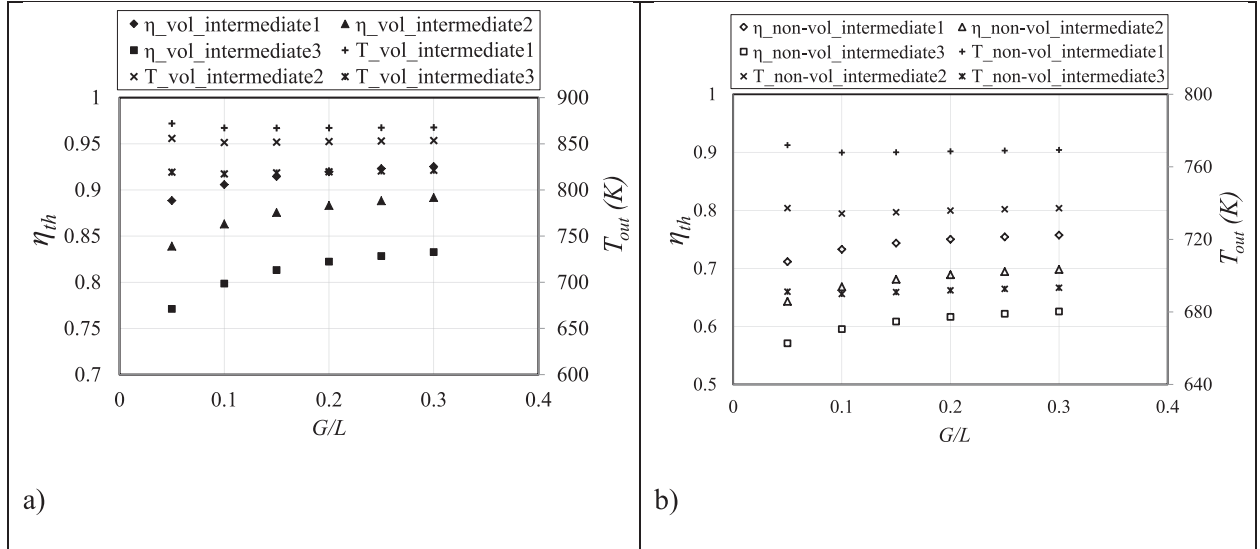


Fig. 27. The effect of G/L on the OVAR thermal efficiency (η_{th}) and the outlet temperature of primary air (T_{out}) with one, two and three intermediate absorber layers, for (a) volumetric heating and (b) non-volumetric heating with $\frac{q}{m_a} = 600$ kJ/kg.

3.3.6. Effect of return air inlet temperature

Fig. 29 depicts the variation of η_{th} and T_{out} for different return air inlet temperatures (300 K – 400 K), different heating conditions, $d_{ab}/L = 1$, $G/L = 0.4$, and $ARR = 0.6$. Following may be inferred:

- η_{th} decreases with the return air inlet temperature for volumetric and non-volumetric heating.
- T_{out} increases with the increasing return air inlet temperature for volumetric and non-volumetric heating.

The above reveals that with the increasing return air inlet temperature, the primary air temperature at the OVAR inlet increases. However, the temperature difference between the absorber and primary air decreases. Consequently, the heat transfer rate decreases, leading to a reduction in heat recovery or thermal efficiency. On the other hand, the increasing values for T_{out} is an out-

come of the overall temperature rise (ΔT) from inlet to outlet. Thus, it can be concluded that ΔT will be lower for a higher return air inlet temperature. This leads to drawing a recommendation that the inlet temperature of return air should be as low as possible for internal cooling for absorbers, thermal re-distribution for the absorbers, and also for achieving high thermal efficiency.

3.3.7. Effect of absorber emissivity

Fig. 30 depicts the variation of η_{th} and T_{out} for different temperature-dependent absorber emissivity [59], different heating conditions, $T_{r,in} = 300$ K, $d_{ab}/L = 1$, $G/L = 0.4$, and $ARR = 0.6$, for one intermediate absorber layer. The maximum relative percentage difference in the computed thermal efficiency (η_{th}) for a wide range ($0.3 \leq \epsilon \leq 0.7$) is around 2–4 %. Also, with increasing value for ϵ the thermal efficiency decreases, as expected, for an OVAR. Similar trend is expected for two and three intermediate absorber

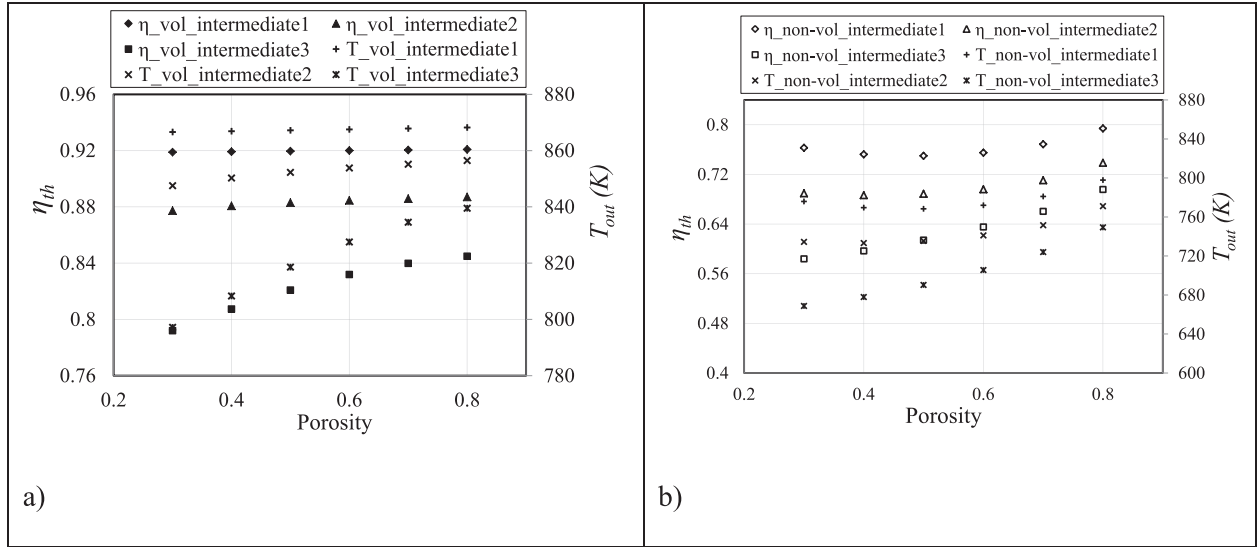


Fig. 28. The effect of absorber porosity on the OVAR thermal efficiency (η_{th}) and the outlet temperature of primary air (T_{out}) with one, two and three intermediate absorber layers, for (a) volumetric heating and (b) non-volumetric heating with $\frac{q}{\dot{m}_a} = 600 \frac{\text{kJ}}{\text{kg}}$.

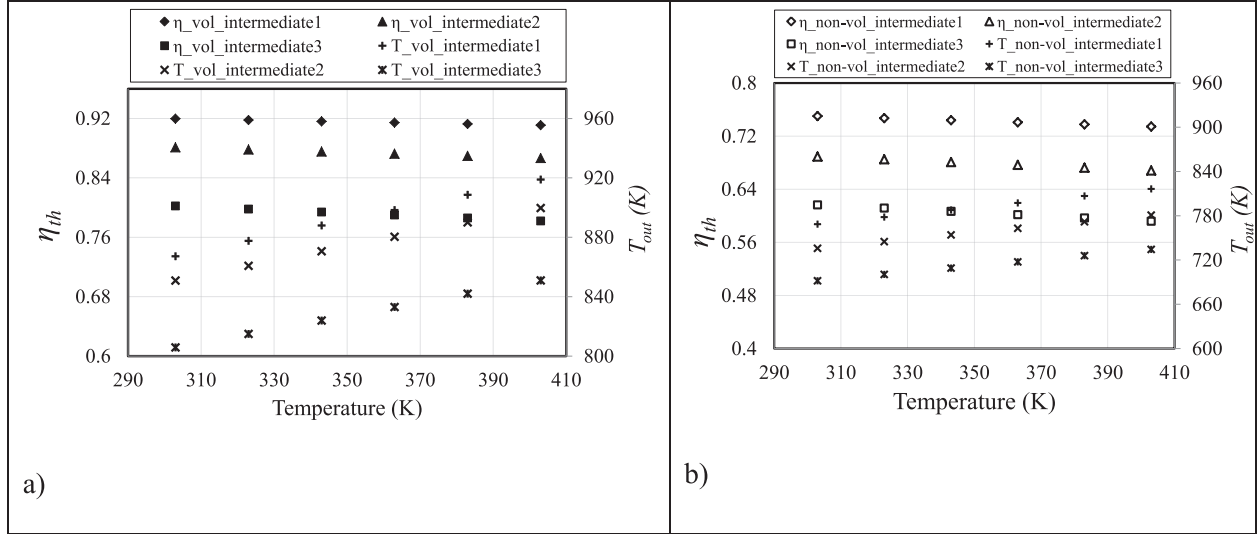


Fig. 29. The effect of return air inlet temperature on the OVAR thermal efficiency (η_{th}) and the outlet temperature of primary air (T_{out}) with one, two and three intermediate absorber layers, for (a) volumetric heating and (b) non-volumetric heating with $\frac{q}{\dot{m}_a} = 600 \frac{\text{kJ}}{\text{kg}}$.

layers. Therefore, the reported values using $\epsilon = 0.6$ are conservative. The unavailability of a suitable correlation between emissivity and porosity for straight pore-based absorbers limits further investigation and remains a prospect for further improvement.

3.3.8. Effect of time-dependent solar irradiance

The effect of time-dependent solar irradiance on the heat transfer in an OVAR is analyzed using the average direct normal irradiance at Jodhpur for July 2012 and December 2012 (see Fig. 31a and 31b) [60]. The computed average air temperatures at the OVAR outlet for the design with one, two and three intermediate layers considering $T_{r,in} = 300 \text{ K}$, $d_{ab}/L = 1$, $G/L = 0.4$, and $ARR = 0.6$ are shown in Fig. 30c and 30d. These reveal the temporal variation and allow operating an application within a certain time interval of a day, depending on the solar irradiance. For example, in the present case, the air temperature exceeds 600 K for July after around 9:00

am, whereas the air temperature exceeds 800 K for December after around 9:00 am.

3.4. Comparative Assessment for the scale-up OVAR designs

So far, straight absorber pore-based scale-up OVAR designs with one, two, and three intermediate absorber layers have been analyzed, and some recommendations are provided for its operation. One final question remains: are the scale-up OVAR designs comparable to the state-of-the-art, in terms of thermal efficiency? To answer this question, further analyses are performed for volumetric and non-volumetric heating with $200 \leq \frac{q}{\dot{m}_a} \leq 800 \frac{\text{kJ}}{\text{kg}}$, $ARR = 0.6$, and the absorber design parameters as in Table 2. For this assessment, the available data from Hoffschmidt et al. [5], Avila-Marin et al. [12], Capuano et al. [61], Singh et al. [44], and Fend et al. [62] are compared with the scale-up OVAR designs in Fig. 32. The following may be inferred:

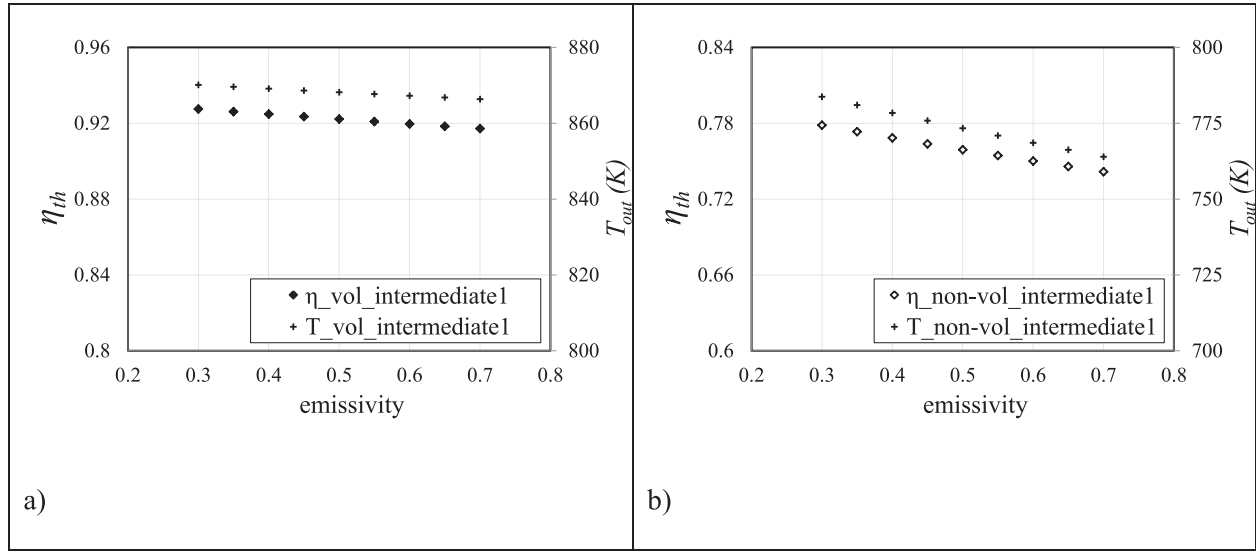


Fig. 30. The effect of emissivity on the OVAR thermal efficiency (η_{th}) and the outlet temperature of primary air (T_{out}) with one intermediate absorber layers, for (a) volumetric heating and (b) non-volumetric heating with $\frac{q}{m_a} = 600 \frac{\text{kJ}}{\text{kg}}$.

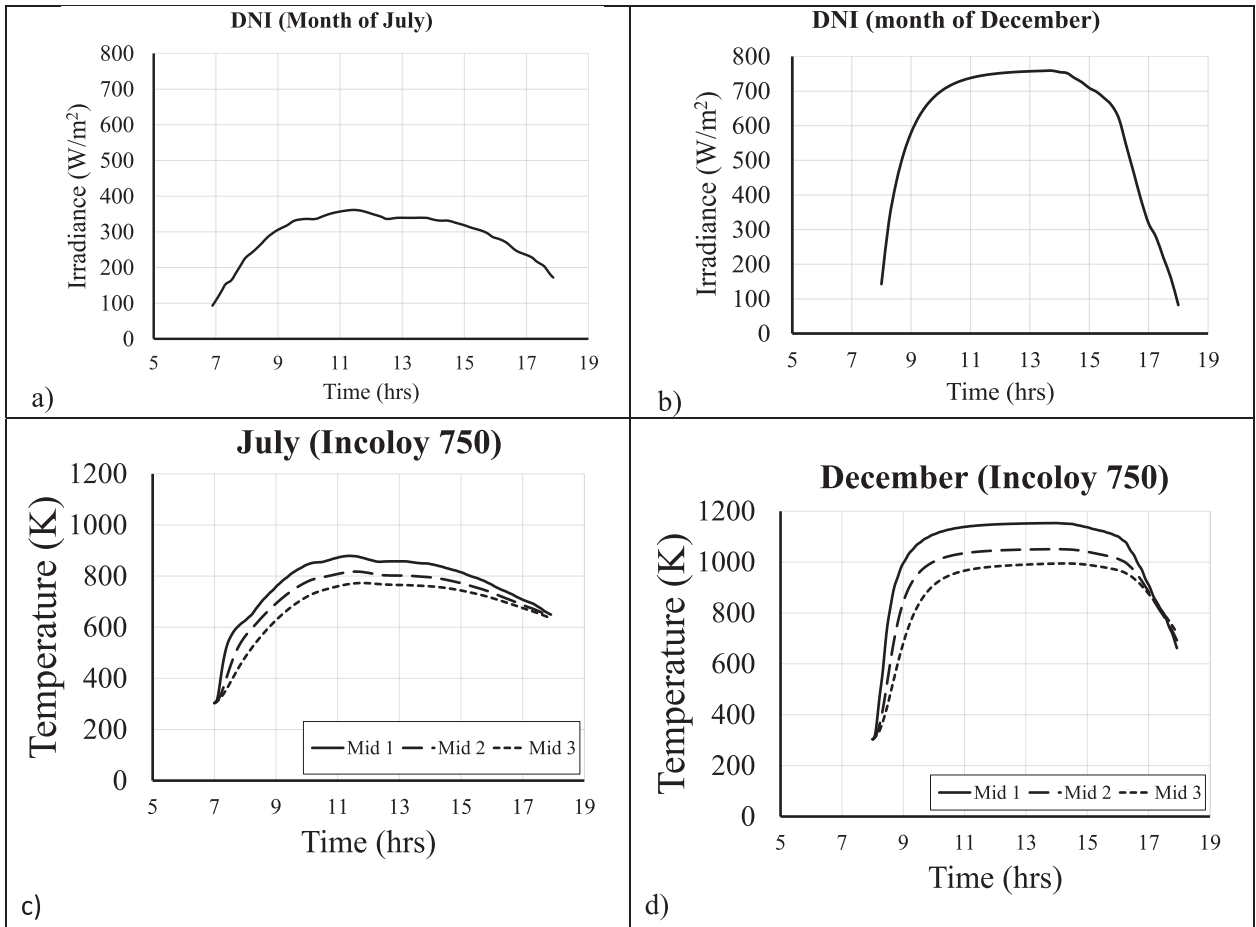


Fig. 31. Time-dependent, monthly averaged, solar irradiance for a) July 2012, b) December 2012 at Jodhpur; the computed time-dependent primary air temperature at the OVAR outlet for c) July 2012 and d) December 2012.

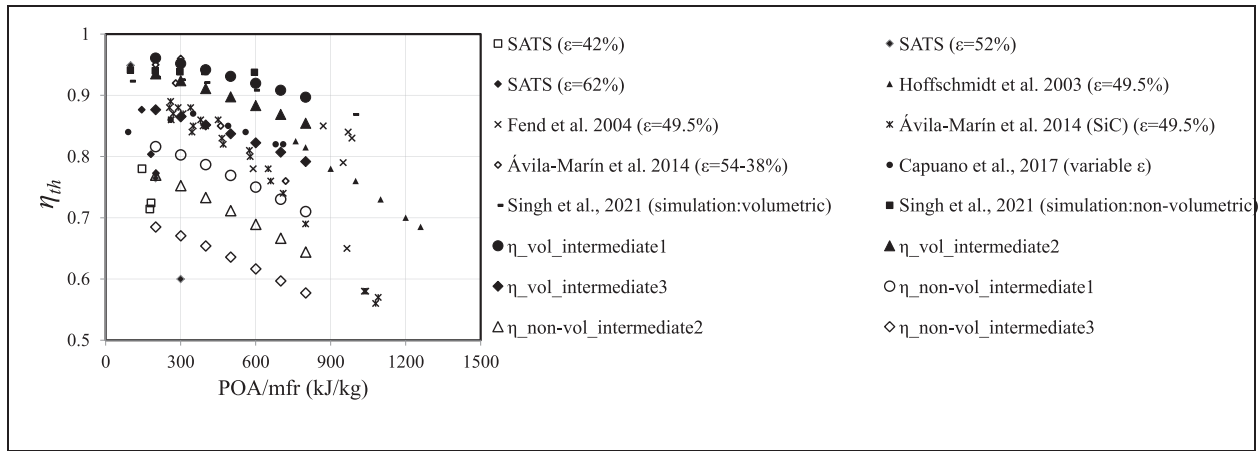


Fig. 32. A comparative assessment of thermal efficiency for the proposed scale-up OVAR design with one, two and three intermediate absorber layers with the reported data.

- with volumetric heating $0.79 \leq \eta_{th} \leq 0.96$ and with non-volumetric heating $0.58 \leq \eta_{th} \leq 0.81$ for the OVAR designs with one, two, and three intermediate absorber layers. This is well in the range of reported data.
- Capuano et al. [61] reported a reasonably high value for η_{th} indicating that the variable porosity-based design offers the volumetric-type absorption. However, such a complex structure will be prone to dust deposition, which is detrimental to its operation at a high temperature [46].
- considering the above, adopting a low exposed surface area for the OVAR designs as in [61] is recommended for the scale-up of straight pore absorber-based OVAR designs, which is also beneficial to mitigate the dust deposition-related critical challenges.
- for volumetric and non-volumetric heating conditions, the recommended ratio of absorber diameter (d_{ab}) to length (L) is $1 \leq d_{ab}/L \leq 2$ for the scale-up OVAR design with three intermediate absorber layers.
- for volumetric and non-volumetric heating conditions, the recommended ratio of absorber gap (G) to length (L) is $0.15 \leq G/L \leq 0.25$ for the scale-up OVAR designs with three intermediate absorber layers.
- based on the performed numerical investigations, the use of multiple intermediate absorber layers is recommended for the scale-up OVAR design. This is expected to mitigate the radial and axial thermal non-uniformity for absorbers and enhance the durability of an OVAR design.

4. Conclusions and recommendations

This paper presents a comprehensive, multi-zone, mathematical model for unsteady heat transfer in a straight pore-based open volumetric air receiver (OVAR). In this model, the porous absorbers are identified as central (C), intermediate (M) and peripheral (P) depending on their locations. In this model, all the modes of heat exchange between absorbers, between absorbers and receiver casing, and between absorbers and return air are included. A comparative assessment between the computed and measured primary air temperature at the absorber outlets revealed that a) the transient and steady phases are reproduced and b) the model predicts the experimental values within an uncertainty of about $\pm 7\%$ at the steady-state. As a next step, the model is used to investigate scale-up OVAR designs with one, two and three intermediate absorber layers, for air mass flow rate $\dot{m}_a = 0.966$ kg/s and input power $\dot{q} = 0.58$ MW or $\frac{\dot{q}}{\dot{m}_a} = 600$ kJ/kg. The analyses lead to the following conclusions and recommendations:

- the temperature difference between absorbers and primary air decreases with the increasing number of intermediate absorber layers for volumetric and non-volumetric heating. The maximum temperature difference between the absorber and primary air at the inlet is less than 400 K for volumetric heating and is greater than 800 K for non-volumetric heating. For volumetric heating, the radiation loss is 1–2 times that of natural convection, and for non-volumetric heating, the radiation loss is 3–4 times that of natural convection. Therefore, the use of a much higher flux concentration is recommended for the non-volumetric heating to attain the desired temperature in comparison to the volumetric heating.
- the recommended air return ratio (ARR) is 0.6, for the scale-up OVAR design, with volumetric and non-volumetric heating conditions. A higher ARR may lead to a reduction of primary air temperature at the absorber outlet.
- with volumetric heating the thermal efficiency, $0.79 \leq \eta_{th} \leq 0.96$ and with non-volumetric heating the thermal efficiency, $0.58 \leq \eta_{th} \leq 0.81$ for the OVAR designs with one, two, and three intermediate absorber layers. These compare well with the existing OVAR designs and the reduction of exposed surface area, following Capuano et al. [61], is recommended for the scale-up OVAR design.
- for volumetric heating, the absorber porosity up to 0.6 is recommended. For non-volumetric heating, a higher value of absorber porosity may be preferred considering absorber material for durability.
- the recommended ratio of (i) absorber gap (G) to length (L) is $0.15 \leq G/L \leq 0.25$ and (ii) absorber diameter (d_{ab}) to length (L) is $1 \leq d_{ab}/L \leq 2$ for the scale-up OVAR design.
- with the increasing inlet temperature of return air the rise in primary air temperature (ΔT) from absorber inlet to outlet decreases. Thus, the thermal efficiency of OVAR decreases with the increasing inlet temperature for return air and is consistent with the reported findings.

Authors' Statement

The corresponding author hereby confirms that the submitted manuscript is original and not submitted elsewhere for publication.

Declaration of Competing Interest

The authors declare that there is no competing interest.

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